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Smart Healthcare Robot

Bringing Care Closer to Patients in Oman

Final Year Project Report

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Introduction

This chapter establishes the project context, problem statement, motivation, objectives, methodology overview, and the project management plan that frames execution for the rest of the report.

1 Introduction

Healthcare in Oman faces a simple but serious challenge: doctors cannot always be where patients need them. This affects everyone - elderly people who cannot travel to hospitals, children in rural areas who need specialist care, and patients in remote communities where the nearest doctor might be hours away. While Oman's healthcare system reaches 100% of urban families, 10% of rural families still lack adequate access to medical care. When doctors are unavailable due to emergencies, travel, or simply being in another location, patients wait. Sometimes they wait too long. This is particularly difficult for elderly patients who may have mobility issues, and for children who need comfort and reassurance during medical situations.[1]

Oman's healthcare system strives to reach all citizens, with government local health services reaching 100% of urban families and 90% of rural families [1], yet geographic distances and resource constraints create barriers to timely and accessible care for the remaining underserved populations. Existing telepresence solutions often rely on manual control and lack the ability to engage patients in even basic diagnostic dialogue. This results in increased workloads for healthcare providers and a less personalized, less effective experience for patients, particularly those in rural areas or with mobility limitations.

Every day in Oman, patients need medical attention that could be provided remotely if the right tools existed. Children in hospitals need comfort when parents cannot be present. Elderly patients need medication reminders and someone to check on their well-being. Rural communities need access to specialist consultations without long journeys.

The global demand for remote healthcare solutions is growing rapidly, with the telepresence robot market expected to experience significant growth. More importantly, Oman's Vision 2040 calls for healthcare innovation that serves all citizens equally, regardless of location [2]. Recent studies have shown that telemedicine adoption in Oman's primary healthcare settings demonstrates both promise and areas for improvement [3], while infrastructure requirements for effective telemedicine networks continue to be refined [4]. The aim of the project is to design and implement a smart telepresence robot with autonomous indoor navigation and on-device basic triage dialogue for ward corridors and bedside spaces in hospitals in Oman.

- To develop a robust autonomous navigation system capable of dynamic obstacle avoidance and Simultaneous Localization and Mapping (SLAM) in crowded hospital corridors.
- To engineer a secure telepresence interface that enables real-time, low-latency audio/visual consultations for remote doctors.
- To integrate an AI-driven interaction layer (using Large Language Models) to provide basic triage, visitor guidance, and patient reassurance.
- To validate the system in a pilot scenario, demonstrating successful autonomous delivery (Nurse Station → Bed) and remote consultation cycles.

The project begins with a systematic requirements analysis informed by a literature review and a healthcare facility needs assessment. Following this, we evaluate and select optimal design alternatives for both the hardware platform and software architecture using the Analytic Hierarchy Process (AHP) methodology. The development follows an iterative prototyping cycle, beginning with establishing reliable remote teleoperation, advancing to implementing and refining the autonomous navigation system, and culminating in integrating the conversational AI dialogue capabilities. Finally, the system undergoes real-world validation in simulated and pilot clinical settings to measure performance metrics such as navigation reliability, telepresence quality of service, and overall user experience for both patients and clinical staff.

The project timeline spans two academic semesters, from September 2025 to May 2026. The first semester focuses on research, requirements analysis, and conceptual design, while the second semester is dedicated to implementation, testing, and final evaluation.

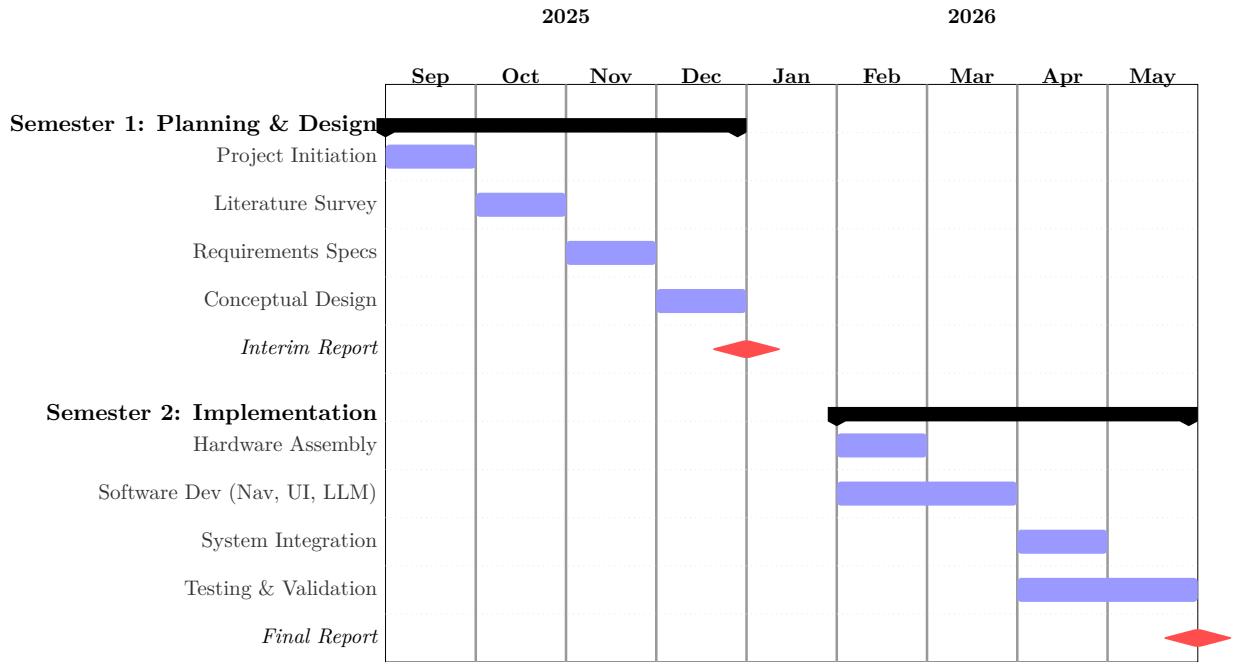


Figure 1: Project Gantt Chart (Sep 2025 - May 2026)

1.0.1 Work Breakdown Structure (WBS)

The detailed tasks and responsibilities are outlined in Table 1.

Table 1: Work Breakdown Structure (WBS)

Phase	Task Description	Deliverables	Assigned To
Phase 1 (Sep 2025)	Project Initiation	Problem Statement	Both
	Methodology Definition	Methodology Section	Both
Phase 2 (Oct 2025)	Literature Review	Literature Survey	Both
	Research Gap Analysis	Gap Identification	Both
Phase 3 (Nov 2025)	Requirements Analysis	Requirements Doc	Both
	Constraints Analysis	Eng. Constraints	Both
Phase 4 (Dec 2025)	Conceptual Design	Design Alternatives	Both
	Functional Decomposition	System Architecture	Both
Phase 5 (Feb-Mar)	Implementation: Hardware	Assembled Robot	Almoulla
	Implementation: Navigation	Autonomous Nav	Almoulla
	Implementation: UI	Clinician/Patient GUI	Abdulmunim
	Implementation: LLM	Voice Interaction	Almoulla
Phase 6 (Apr-May)	Testing & Finalization	Integrated System	Both
	Final Report Writing	Final Thesis	Both

2

Literature Review

This chapter surveys consumer, industrial, and healthcare robotics; documents benefits and challenges of AMRs in hospitals; analyzes core technologies (navigation, RF localization, SLAM, LLMs); and motivates the platform choices and project needs.

2 Literature Review

2.1 Motivation and State of the Field

The integration of robotics and artificial intelligence into healthcare is advancing rapidly, driven by challenges such as aging populations, rising healthcare costs, and medical staff shortages. Researchers have developed various robotic solutions, including telepresence robots for remote consultations, autonomous robots for logistics, and socially assistive robots (SARs) for patient engagement. While each type has proven useful, they often function in isolation, addressing only part of patients' and clinicians' overall needs. The goal is to show that although the components for such a system already exist, combining them into a single, user-friendly, and autonomous platform remains an underexplored area with high potential, especially for improving healthcare accessibility in regions like Oman.

Telepresence robots, often described as a mobile screen, are widely used for remote medical consultations. Systems such as InTouch Health RP-VITA and the VGo robot have been used successfully in "telestroke" care, allowing neurologists to assess patients remotely, reduce diagnosis time, and improve outcomes [5]. These systems support remote rounds, specialist consultations, and family visits, cutting travel costs and reducing infection risks. However, studies also note several drawbacks. A review by Kristoffersson et al. found that telepresence robots, though functional, often feel impersonal and difficult to control [6]. Clinicians report high cognitive load when navigating cluttered hospital spaces. Patients, especially the elderly and children, also find these robots less engaging than in-person visits due to limited physical embodiment and lack of expressive cues [7]. This "presence gap" limits their effectiveness beyond basic video conferencing, reducing their ability to offer comfort and reassurance during care.

This literature review builds a comprehensive case for an integrated, autonomous healthcare robotics platform. It begins by examining success stories from consumer and industrial robotics, demonstrating that autonomous navigation technology has reached maturity and widespread adoption. The review then analyzes healthcare-specific applications, including telepresence, socially assistive robots, and hospital logistics systems, alongside their underlying technical components such as navigation systems, mobile platforms, conversational AI, and user interfaces. By systematically evaluating these technologies and their implementations, this survey identifies a critical research gap: the absence of a unified system that synthesizes clinical telepresence, social interaction, and autonomous mobility into a single platform.

2.1.1 Success Stories: Consumer, Industrial, and Quadruped Robotics

1. Consumer Market: The Robotic Vacuum Revolution.

The most successful deployment of autonomous mobile robots to the general public has been robotic vacuum cleaners, which dominate the consumer robotics market without serious competition from traditional cleaning methods. The global robotic vacuum cleaner market has experienced explosive growth, with projections indicating expansion from \$9.07 billion in 2024 to \$31.79 billion by 2033, representing a compound annual growth rate (CAGR) of 16.97% [8]. Global shipments reached 15.35 million units in the first half of recent years, demonstrating a 33% year-over-year growth [9].



Figure 2: Modern robotic vacuum cleaners utilize SLAM and localization technologies to autonomously navigate residential spaces, representing the most successful consumer robotics application to date.

The key to their success lies in the sophisticated implementation of SLAM (Simultaneous Localization and Mapping) and localization technologies. Modern robotic vacuum cleaners employ LiDAR sensors (Light Detection and Ranging), visual SLAM (VSLAM), and advanced mapping algorithms to create real-time maps of residential environments, enabling efficient path planning and obstacle avoidance [10]. This technology has proven so reliable that robotic vacuums have achieved nearly 40% of the service robot market share [11], demonstrating that autonomous navigation, when properly implemented, is mature enough for widespread consumer adoption.

2. Industrial Market: Amazon’s Warehouse Robotics Dominance.

In the industrial sector, the most successful large-scale application of AMRs (Autonomous Mobile Robots) has been warehouse automation, with Amazon leading the field. Amazon reports more than 750,000 to over one million robots across its fulfillment network, including drive units, mobile manipulators, and next-generation platforms such as Pegasus and Xanthus [12, 13]. These fleets demonstrate reliable fleet management, safe human-robot collaboration, and 24/7 uptime in highly dynamic environments.



Figure 3: Amazon Pegasus drive unit used in high-throughput fulfillment centers.

3. Quadruped Robotics: Unitree Go2 as a Robust Mobility Platform.

State-of-the-art quadruped robots such as Unitree Go2 demonstrate robust mobility on unpredictable terrain using reinforcement learning for locomotion policy optimization, advanced multi-sensor perception (3D LiDAR + depth/RGB cameras), and high-performance

onboard compute for real-time planning and control. These capabilities enable operation on stairs, slopes, and uneven surfaces, with compelling applications in inspection, security, and emergency response [14, 15]. Although not our primary target platform, Unitree’s capabilities illustrate complementary directions for rugged environments using RL and advanced multi-sensor perception, and inspire future extensions.



Figure 4: Unitree Go2: quadruped platform featuring RL locomotion, multi-sensor perception, and high-performance compute.

2.1.2 Benefits of Autonomous Mobile Robots in Healthcare Settings

Recent literature and deployment data reveal multiple evidence-based benefits of AMRs in hospital environments:

Infection Control and Reduced Human Contact. AMRs significantly minimize human-to-human contact, reducing cross-contamination risks a critical advantage demonstrated during the COVID-19 pandemic [16, 17]. Autonomous delivery robots eliminate the need for multiple staff members to handle potentially contaminated materials, reducing occupational exposure and conserving personal protective equipment. UV-C (ultraviolet-C) disinfection robots have been shown to achieve 99.9% pathogen kill rates on environmental surfaces, helping reduce hospital-acquired infections (HAIs) [18].

Wayfinding and Visitor Guidance. Intelligent hospital guidance robots provide accurate navigation assistance to visitors and patients, reducing confusion and staff burden [19]. These robots can escort visitors to patient rooms, provide multilingual directions, and answer frequently asked questions about hospital facilities, allowing staff to focus on clinical care.

Material Transport and Delivery. AMRs excel at routine delivery tasks including medications, meals, linens, laboratory specimens, and medical supplies. This automation frees nursing staff from non-clinical transport duties, allowing them to spend more time on direct patient care. Studies indicate that delivery robots can reduce staff time spent on logistics by up to 30% [20].

Psychological Support and Patient Morale. Research consistently demonstrates that pediatric patients exhibit positive emotional responses to robot interactions [21]. Socially interactive robots reduce anxiety, provide distraction during painful procedures, and improve treatment compliance in children. Studies show that patients perceive robotic companions as non-judgemental, reducing psychological barriers to seeking help or expressing concerns [22].

Autonomous Disinfection and Cleaning. Cleaning and disinfection robots can be equipped with UV-C light systems at relatively low cost, providing continuous environmental sanitation [16]. These robots operate during off-peak hours without human supervision, ensuring consistent hygiene standards while minimizing disruption to clinical workflows.

Telepresence and Virtual Presence. AMRs with integrated telepresence capabilities enable remote consultations, family visits, and specialist rounds without requiring travel [5]. This technology proved invaluable during pandemic restrictions and continues to provide value for rural healthcare access and specialist consultations.

Waste Collection and Handling. Some advanced AMR systems are deployed for biohazardous waste collection and disposal, reducing staff exposure to potentially infectious materials [20]. In China and other Asian markets, robots have been successfully implemented for collecting and transporting medical waste from patient rooms to centralized disposal areas, demonstrating significant improvements in operational efficiency and staff safety.

2.1.3 Challenges in Hospital AMR Deployment

Despite the significant benefits, several challenges remain:

Cost and Return on Investment. Hospital-grade AMRs represent substantial capital investments, typically ranging from \$50,000 to \$150,000 per unit depending on capabilities. Healthcare administrators require clear Return on Investment (ROI) demonstration through labor savings, efficiency gains, and improved outcomes to justify procurement [20].

System Integration Complexity. Integrating AMRs with existing hospital information systems (HIS), electronic health records (EHR), elevator controls, automatic doors, and staff communication systems presents significant technical challenges. Interoperability standards are still evolving, requiring custom integration work for each deployment [17].

Safety, Privacy, and Trust. Regulatory compliance (HIPAA, GDPR), patient data security, and physical safety in crowded environments remain critical concerns. Healthcare facilities must ensure that robots can safely navigate around patients, visitors, and medical equipment while maintaining patient privacy and data security [23]. Building trust among staff and patients requires careful change management and demonstration of reliability.

Regulatory and Liability Issues. Medical device regulations, liability frameworks for autonomous systems, and insurance considerations create additional barriers to widespread adoption. Healthcare organizations must navigate complex approval processes and establish clear accountability frameworks for robot operations.

2.1.4 Overview of Existing Robots in Healthcare

The hospital robotics market has matured significantly, with several manufacturers deploying autonomous mobile robots at scale. Table 2 presents a comprehensive comparison of leading

hospital robot platforms currently deployed in eastern markets. Note that prices are approximate estimates and can vary significantly depending on region, configuration, volume, and additional features.

Rank	Manufacturer & Model	Est. Deploy-ments	Est. Price (USD)	Key Hospital Applications	Features & Notes
1	Pudu Robotics – PuduBot 2 [24]	15,000–20,000 units (5,000 healthcare)	\$12,000	Medication/meal delivery, linen transport, waste collection	Multi-floor navigation (elevator integration), VSLAM+LiDAR, 40 kg payload, 6–8 hr battery. Deployed in 100+ hospitals globally (China, Singapore, U.S.); customizable trays, infection-resistant coating.
2	Keenon Robotics – M2 Medical Robot [25]	8,000–10,000 units (healthcare-focused)	\$47,500	UV-C disinfection, medication/lab sample delivery, patient room cleaning	UV sterilization (99.9% pathogen kill rate), 40 kg payload, voice interaction, 10 hr runtime. Used in 100+ Chinese hospitals during COVID; exported to U.S./Europe (HIPAA-compliant).
3	Pudu Robotics – PUDU CC1 [26]	5,000–7,000 units (healthcare + hospitality)	\$16,800	Floor cleaning, disinfection, material transport	Autonomous scrubbing (1,500 m ² /hr), self-charging, obstacle avoidance. Popular in Chinese/U.S. hospitals for large-scale sanitation; integrates with PuduBot for hybrid tasks.
4	Keenon Robotics – Peanut Medical [27]	3,000–5,000 units	\$12,700	Medication/meal delivery, patient guidance, telepresence	Compact (59 cm wide), multi-modal navigation, 30 kg payload. Deployed in Asia-Pacific hospitals; supports remote doctor-patient interaction via screen.

Table 2: Competitive analysis of leading hospital AMR platforms by global deployment and capabilities.



Figure 5: Pudu Robotics PuduBot 2



Figure 6: Keenon M2 Medical Robot



Figure 7: Pudu Robotics PUDU CC1



Figure 8: Keenon Peanut Medical

In parallel, the field of Socially Assistive Robotics (SAR) has focused on the psycho-social aspects of patient care. These robots are designed not to perform physical tasks but to engage, encourage, and comfort patients through social interaction. For pediatric patients, robots like NAO and PARO (a therapeutic baby seal robot) have been shown to significantly reduce pain perception and anxiety during painful medical procedures such as vaccinations and chemotherapy [21]. The robot acts as a coach and a distractor, creating a more positive and cooperative atmosphere. For elderly populations, SARs have been explored for companionship to combat loneliness, provide cognitive stimulation for dementia patients, and deliver medication reminders to improve treatment adherence [28]. These studies confirm that a robot’s ability to interact socially is a powerful tool for improving patient well-being and clinical outcomes. The primary limitation of SARs, however, is that they are typically specialized, non-mobile or semi-autonomous platforms that lack the high-fidelity telepresence capabilities needed for direct clinical consultation.

2.2 Technical Building Blocks

This section deconstructs the essential technologies that form the foundation of an advanced autonomous healthcare robot. It examines navigation systems including Radio Frequency (RF) localization and SLAM-based sensor fusion, mobile robot platform specifications and selection criteria, Conversational AI and Large Language Models (LLMs) for intelligent interaction, and User Interface (UI) and Human-Robot Interaction (HRI) design paradigms for clinical integration.

2.2.1 Navigation System

The challenge of manual navigation highlighted in telepresence research is being addressed by autonomous mobile robots (AMRs). Modern AMRs employ a combination of SLAM, path planning, and obstacle avoidance to navigate complex environments without human intervention. The core navigation stack typically includes Radio Frequency (RF) Localization Technologies and SLAM-Based Sensor Fusion Techniques.

Radio Frequency (RF) Localization Technologies Radio frequency (RF) refers to electromagnetic waves (3 kHz–300 GHz) used for wireless communication. RF-based localization is essential where GPS is unreliable, such as indoors or through obstacles. Technologies like Wi-Fi, RFID, Bluetooth Low Energy (BLE), ZigBee, and Ultra-Wideband (UWB) are used, each selected based on coverage, scalability, battery life, accuracy, and cost. UWB, for example, enables multi-node scalability and multi-year tag battery life, but at higher cost (e.g., up to \$20,000 for a 5000 m² facility). Accuracy ranges from about 20 cm to over 1 m, with performance degrading in cluttered or metallic environments [29]. The general structure of these systems is to use static reference sensor locations, or anchors, within an environment, along with beacons or tags fixed to the objects with desired tracking. The location of the tracked object can be estimated using readings from the anchors. Mathematically, this estimation is commonly called multilateration (also called trilateration) when using distance measurements. Figure 9 shows an example UWB anchor-tag geometry, while figure 10 illustrates the multilateration technique.



Figure 9: UWB anchor-tag geometry for ranging and multilateration-based localization in indoor environments.

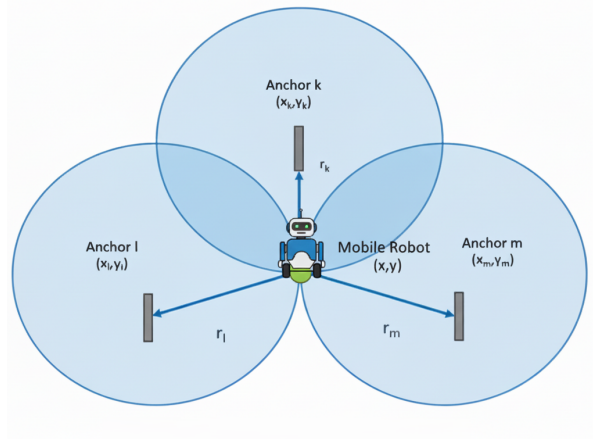


Figure 10: Illustration of multilateration technique. Each known anchor forms a circle with radius equal to its distance to the robot, and the robot's estimated position is where these circles intersect.

Angle of Arrival (AoA) is a localization technique that determines the direction or angle from which a radio signal is received. An antenna array senses time differences across elements and applies signal-processing algorithms such as MUSIC or ESPRIT to convert those differences into an angle. AoA (often via BLE 5.1+ or UWB Direction Finding) estimates a target's angle using a beacon or tag and a robot-mounted antenna array, reducing reliance on dense static infrastructure. BLE-based tags offer excellent battery life (years). Limitations include line-of-sight sensitivity and variable accuracy; hybrid AoA + RSSI filtering can improve precision, achieving mean errors on the order of 12 cm in simulation even in cluttered settings [30]. AoA could be used to implement a solution that does not depend on localization or positioning infrastructure in the environment, instead having the robot home in on a beacon worn by the

user. This would be similar to how some Bluetooth trackers work for finding lost items. The robot would use the angle of arrival data to determine the direction to move towards the beacon signal, allowing it to autonomously navigate to the user’s location. This approach could be particularly useful in dynamic environments like hospitals where fixed infrastructure may not be feasible. Figure 11 illustrates a proposed setup where a Bluetooth homing controller on the robot uses AoA data from a wearable beacon to guide navigation towards the target. Multiple anchors could be used to implement waypoints following navigation paths through the environment.



Figure 11: Configuration of proposed setup of Bluetooth homing controller.

SLAM-Based Sensor Fusion Techniques Simultaneous Localization and Mapping (SLAM) is a critical technology for autonomous navigation, enabling robots to build maps of unknown environments while simultaneously keeping track of their own location. Its effectiveness hinges on robust sensor fusion techniques that integrate data from various sources such as camera(s), IMU, wheel odometry, or RF ranging. Beyond external RF systems, autonomous navigation heavily relies on SLAM with multi-sensor fusion for robustness. A common stack fuses 2D LiDAR with an IMU and wheel odometry. IMU and odometry constrain motion, correcting LiDAR motion distortion and mitigating slip-induced drift. Studies report substantial gains when integrating IMU and wheel odometry, for example, approximately 27% improvement in rotation error and greater than 67% reduction in position error relative to LiDAR only SLAM, at the expense of added system complexity and compute [31].

VSLAM (Visual Simultaneous Localization and Mapping) is a method that uses cameras to help robots figure out where they are and build maps of their surroundings at the same time. VSLAM uses cameras like monocular (single), stereo (two), or RGB-D (depth-sensing) for positioning and mapping. It provides low-cost, lightweight hardware and better scene understanding. Challenges include poor performance in texture-less or repetitive areas, and scale issues in monocular systems (e.g., early ORB-SLAM 2.0). Modern approaches prefer hybrid Visual-Inertial-Ranging-LiDAR (VIRAL) SLAM for improved reliability in complex environments [32]. Figure 12 shows the combination of visual/laser data with inertial and odometric inputs for stable estimation. Tools like Google Cartographer build 3D maps in real-time, using LiDAR, IMU, and odometry.

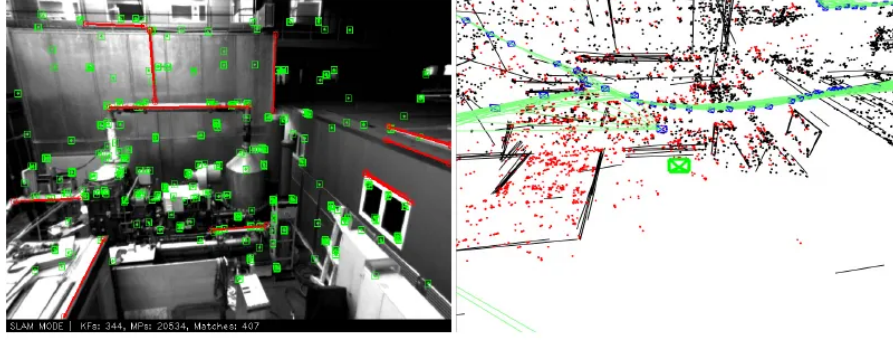


Figure 12: Illustration of SLAM fusion: visual/laser perception integrated with inertial and odometric constraints for robust state estimation.

Navigation Systems Across Hospital Robotics Platforms. Table 3 summarizes the navigation technologies and sensor configurations employed by leading hospital robotics platforms mentioned in table 2.

Robot Model	Primary Localization	Mapping Technology	Key Sensors
PuduBot 2 [24]	SLAM (LiDAR-based)	Real-time 2D/3D mapping	2D LiDAR, depth camera, IMU
M2 Medical Robot [25]	SLAM (LiDAR-based)	Real-time 2D mapping	2D LiDAR, RGB camera, encoders
PUDU CC1 [26]	SLAM (LiDAR-based)	Real-time 2D mapping	2D LiDAR, ultrasonic sensors, encoders
Peanut Medical [27]	Hybrid (SLAM + RF)	Real-time 2D mapping	LiDAR, depth sensors, optional UWB

Table 3: Navigation systems and sensor technologies across leading hospital robotics platforms.

2.2.2 Mobile Robot Platform specifications

The selection of an appropriate mobile robot platform is a critical decision that directly impacts system reliability, development time, and operational costs. For a healthcare telepresence robot operating in hospital corridors and patient rooms, the platform must satisfy several essential technical and operational requirements while remaining within practical budget constraints.

Platform Requirements. The system requires a wheeled mobile robot platform with the following capabilities:

- **Autonomous docking and charging** with high-level API commands (`dock()`, `return_to_charge()`)
- **ROS/ROS 2 compatibility** for navigation packages (Nav2, SLAM Toolbox, AMCL)
- **Payload capacity** of 5–10 kg for telepresence screen, onboard computer, and camera system
- **Indoor navigation sensors** including LiDAR (2D or 3D) and depth cameras
- **Safety and reliability** features (cliff detection, bumpers, emergency stop)

Project Constraints. Budget constraints significantly influence platform selection. The target budget for the mobile base, charging dock, and core sensors is approximately \$1,000–\$1,500 USD, recognizing that higher-cost industrial platforms (\$5,000+) exceed the scope of an educational research project. Additional constraints include global availability (shipping to

Oman), vendor support and documentation quality, and community resources for troubleshooting and development.

Platform Comparison. Table 4 presents a comparative analysis of commercially available mobile robot platforms.

Platform	Price (USD)	Dock	ROS	Payload	Notes
<i>Budget Tier (\$500–\$1,000)</i>					
iRobot Create 3 [33]	\$549	Yes	ROS 2	9 kg	Roomba-based, proven docking, ideal for education
Yahboom ROSMAS-TER X3 [34]	\$699	DIY	ROS 1	5–8 kg	Mecanum wheels, requires custom docking solution
Elephant Robotics MyAGV 2023 [35]	\$899	DIY	ROS 1	8 kg	Mecanum wheels, 2D SLAM ready, Python/ROS compatible
<i>Mid-Tier (\$1,000–\$2,000)</i>					
SLAMTEC Apollo [36]	\$1,200	Yes	ROS 1/2	50 kg	Same platform as Pudu/Keenon hospital robots
TurtleBot 4 Standard [37]	\$1,400	Yes	ROS 2	9 kg	Research-grade, Nav2 pre-configured, Lidar included
Husarion ROSbot 2R [38]	\$1,600	Optional	ROS 2	10 kg	Mecanum wheels, hospital navigation demos available
AgileX LIMO Pro [39]	\$1,799	Optional	ROS 2	10 kg	4-mode steering, Jetson Nano, research platform

Table 4: Comparative analysis of mobile robot platforms for healthcare telepresence applications.

2.2.3 Conversational AI and Large Language Models (LLMs)

The field of large language models (LLMs) has witnessed transformative advancements, establishing them as a core technology for conversational AI. An LLM is a sophisticated artificial intelligence system trained on vast datasets of text and code, enabling it to understand, generate, and interact with human language in a nuanced manner. These models have rapidly transitioned from research prototypes to industrial-scale technologies, resulting in a proliferation of models with distinct strengths and commercial terms.

The LLM landscape is broadly divided into two categories: open-source and proprietary. Open-source models, such as Meta’s Llama series and the DeepSeek series, offer significant flexibility, allowing for deep customization and fine-tuning to meet specific application needs, often at a lower cost [40]. However, they may require more complex setup and can sometimes lag behind proprietary counterparts in raw performance. In contrast, privately-owned models like OpenAI’s GPT series and Google’s Gemini series typically offer state-of-the-art performance and are adept at handling highly complex tasks. This performance comes with trade-offs, including higher operational costs and potential data privacy concerns, which are critical considerations in healthcare applications [41]. The selection of an LLM depends on a careful balance between performance requirements, budget, customizability, and data governance policies. Table 5 provides a comparative overview of leading LLM options.

Model (Version)	Org.	License	Max Context (tokens)	Input API Cost (\$/M)	Output API Cost (\$/M)
OpenAI GPT-5.2 [42]	OpenAI	Proprietary	128,000	\$1.75	\$14.00
Google Gemini 3 Flash [43]	Google	Proprietary	1,000,000	\$0.50	\$3.00
Google Gemini 2.5 Pro [43]	Google	Proprietary	1,000,000	\$1.25–\$2.50	\$10.00–\$15.00
Google Med-Gemma 2B	Google	Open Weight (Apache 2.0)	8,192	\$0.00 (Self-hosted)	\$0.00 (Self-hosted)
Meta Llama 4 Maverick [44]	Meta	Llama Community License	10,000,000	\$0.19–\$0.49	\$0.19–\$0.49
DeepSeek V3.1 [45]	DeepSeek	MIT (Code) / DeepSeek Model	128,000	\$0.14	\$0.28

Table 5: Comparative analysis of leading Large Language Models with December 2025 pricing and specifications.

LLM Discussion and Cost Considerations. Table 5 summarizes key characteristics of leading LLM options as of December 2025. OpenAI’s GPT-5.2 offers strong performance with a 128k token context window at \$1.75/\$14.00 per million input/output tokens. Google’s Gemini 3 Flash provides exceptional value with a 1M token context and competitive pricing at \$0.50/\$3.00, with free tier available for development (up to 500 requests per day). Google’s Gemini 2.5 Pro offers higher capabilities for complex tasks at \$1.25–\$2.50 input and \$10.00–\$15.00 output per million tokens. Google’s Med-Gemma represents a family of specialized open-weight models (available in 2B, 7B, and 27B parameter variants) fine-tuned specifically on medical literature, clinical guidelines, and healthcare datasets [46]. While Med-Gemma demonstrates state-of-the-art performance on medical benchmarks (MedQA, MedMCQA, PubMedQA) and would be highly relevant for advanced clinical decision support or medical question answering, it is considered overkill for the basic conversational interactions required in this project (patient guidance, wayfinding, simple information queries). Meta’s Llama 4 Maverick stands out with an unprecedented 10M token context window and industry-leading cost efficiency (\$0.19–\$0.49 per million tokens), making it highly suitable for long-context healthcare applications. DeepSeek V3.1 offers the most economical option at \$0.14/\$0.28 per million tokens, particularly beneficial for cost-sensitive deployments. For this project’s conversational requirements (patient guidance, wayfinding, simple information queries), either Gemini 3 Flash (free tier) or DeepSeek V3.1 (lowest cost) would be appropriate choices, with Llama 4 Maverick offering future-proof long-context capabilities if advanced document processing becomes necessary.

2.2.4 User Interface and Human-Robot Interaction

For autonomous healthcare robots operating in clinical environments, the user interface (UI) and human-robot interaction (HRI) design directly impacts adoption, user satisfaction, and clinical workflow integration. The interface must balance accessibility for diverse user groups (clinicians, administrative staff, patients) with robust control mechanisms suitable for dynamic hospital environments. Common UI paradigms employed in hospital robotics include:

Onboard Touchscreen Interface. A dedicated touchscreen mounted on the robot body provides direct, localized control and real-time status feedback. Common in systems like Pudu Robotics’ PuduBot 2 and Keenon Robotics’ M2 Medical Robot, these interfaces typically run

Android OS and display navigation status, task queues, and emergency controls. Advantages include intuitive interaction without external devices and immediate feedback. Limitations include reliance on the robot’s physical proximity and potential hygiene concerns in sterile environments [25].

Mobile Application (iOS/Android). Native mobile applications running on clinicians’ and administrators’ smartphones or tablets enable remote operation and fleet monitoring from anywhere within hospital networks. This approach is ubiquitous across commercial systems (PuduBot 2, M2, Peanut Medical) and supports features such as route customization, real-time tracking, task scheduling, and status notifications. Mobile apps offer flexibility and integrate seamlessly with existing hospital workflows, though they require network connectivity and depend on user device management [24].

Web-Based Dashboard. Cloud-hosted or on-premises web portals provide centralized fleet management, historical analytics, and administrative controls for hospital operators and IT administrators. Systems like Keenon Robotics and Pudu Robotics include web dashboards for route optimization, performance metrics, and multi-robot coordination. Web interfaces support remote monitoring across multiple locations and facilitate integration with hospital information systems (HIS). Trade-offs include network dependency, data security considerations, and potential latency in real-time control [26].

Voice Control and Natural Language Interfaces. Voice-based interaction leveraging conversational AI and LLMs (as discussed in Section 2.2.3) represents an emerging paradigm for hands-free operation in clinical settings. Systems like Keenon’s M2 support voice commands, allowing clinicians to interact with robots while maintaining hygiene and workflow continuity. Voice interfaces reduce cognitive load and support more natural, intuitive interactions, but require robust speech recognition, noise handling (hospital environments are loud), and contextual understanding [41].

Hospital Integration via APIs. Integration with existing hospital IT infrastructure through RESTful APIs or DICOM/HL7 standards allows robots to receive task assignments directly from electronic health records (EHR) or hospital management systems. This approach minimizes manual input and ensures tasks are coordinated with clinical workflows. Examples include systems capable of receiving delivery requests directly from hospital pharmacies or labs [26].

Robot UI Usage Across Hospital Platforms. Table 6 summarizes the UI and interaction modalities employed by leading hospital robotics platforms, illustrating the diversity of approaches and trade-offs in current systems.

Robot Model	Onboard Screen	Mobile App	Web Dash-board	Voice/AI In-terface
PuduBot 2 [24]	Android tablet	iOS/Android	Yes	Limited
M2 Medical Robot [25]	Android display	iOS/Android	Yes	Yes (voice control)
PUDU CC1 [26]	Android panel	iOS/Android	Yes	No
Peanut Medical [27]	Optional	iOS/Android	Yes	Limited

Table 6: UI and interaction modalities across leading hospital robotics platforms.

2.3 Conclusion

The literature presents a clear picture:

1. **Telepresence robots** are effective for remote diagnosis but are limited by manual control and an impersonal user experience.
2. **Socially assistive robots** have proven their value in patient engagement and comfort but lack clinical telepresence and advanced mobility.
3. **Autonomous mobile robots** have mastered navigation in hospitals but are used for logistics, not patient interaction.
4. **Conversational AI** offers intelligent dialogue but lacks the physical embodiment needed for meaningful healthcare encounters.
5. **Consumer and industrial robotics** have demonstrated that autonomous navigation technology is mature, reliable, and cost-effective for real-world deployment, as evidenced by millions of robotic vacuum cleaners in homes and over one million AMRs in Amazon warehouses.
6. **Current hospital AMRs** are predominantly focused on material transport and disinfection, with limited integration of social interaction, advanced telepresence, and conversational AI capabilities.

The critical research gap lies at the intersection of these domains. There is a demonstrable need for a single, integrated platform that combines the **clinical access** of a telepresence robot, the **patient-centric engagement** of a SAR, the **operational independence** of an autonomous navigator, and the **interactive intelligence** of modern conversational AI. The proven success of autonomous navigation in consumer (robotic vacuums) and industrial (Amazon warehouses) applications demonstrates that the core technology is mature and ready for advanced healthcare applications.

This project directly addresses this gap by designing a holistic system that synthesizes these strengths, creating a smart healthcare robot capable of not only connecting doctors and patients but also navigating its environment independently and engaging with patients in a helpful and empathetic manner. By learning from the successes of consumer robotics, warehouse automation, and existing hospital AMR deployments, this project aims to advance the state-of-the-art in healthcare robotics toward truly integrated, intelligent, and patient-centered care.

3

System Conceptual Design

This chapter presents the high-level system architecture of the smart healthcare robot, including the overall system overview, design alternatives, and justification for the selected hardware and software configuration.

3 System Conceptual Design

3.1 Introduction: System Overview

The proposed smart healthcare robot is a mobile, autonomous platform designed to assist hospital staff in routine, non-critical tasks while providing patients with an intuitive human–robot interaction (HRI) interface. The robot operates primarily in indoor clinical environments such as hospital wards, corridors, and waiting areas. Its main roles include autonomous navigation, safe obstacle avoidance, point-to-point delivery of small items, basic patient interaction via conversational AI, and remote telepresence capabilities for clinical staff.

From a functional perspective, the robot must be able to localize itself in the hospital environment, plan safe paths between waypoints, and react to dynamic obstacles such as people, trolleys, and doors. It should expose a user-friendly interface through a tablet and voice interaction for nurses, doctors, and patients to issue tasks or receive information. The system is conceived as a modular platform so that sensors, software modules, and interaction modalities can be extended in future iterations without major redesign.

The overall system architecture follows a layered approach consisting of:

- **Mobile Base Layer:** Providing locomotion, payload mounting, and physical robustness
- **Computing and Control Layer:** High-level computer running ROS 2 and microcontroller for real-time motor control
- **Navigation and Perception Layer:** Sensors for localization, mapping, and obstacle detection
- **Intelligence Layer:** Large Language Model (LLM) for natural language understanding and HRI
- **Communication and Interface Layer:** Tablet UI, web dashboard, and voice interaction
- **Power Management Layer:** Battery system with automatic docking capabilities

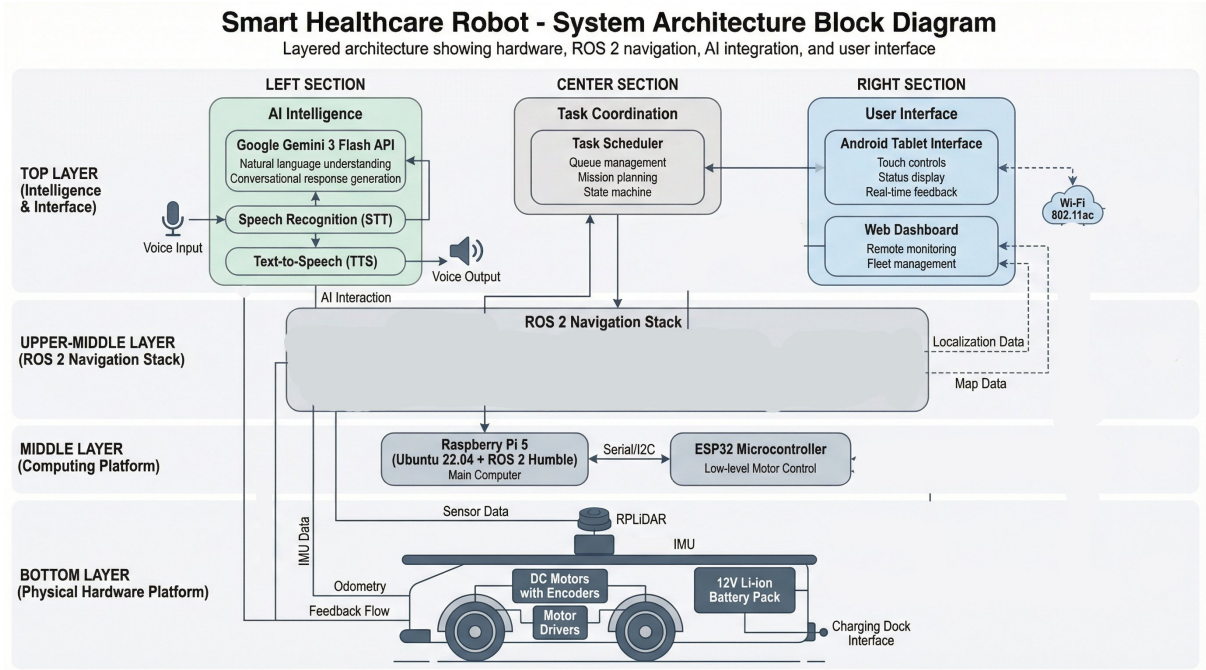


Figure 13: System architecture block diagram showing the layered design approach: physical hardware platform (bottom), computing and control layer, ROS 2 navigation stack, and intelligence/interface layer (top). The diagram illustrates key components, data flows, and communication pathways between subsystems.

This conceptual design aims to balance technical performance, patient safety, modularity, and cost, providing a realistic path from prototype to potential deployment in Omani hospitals.

3.2 Design Alternatives and Selection

To arrive at a suitable system architecture for the smart healthcare robot, three main design alternatives were systematically evaluated. The evaluation focuses on robot platform approach, computing architecture, sensor configuration, and AI integration capabilities.

3.2.1 Alternative A: Commercial Pre-Built Platform

This alternative leverages an existing commercial robot platform designed for healthcare or service applications, such as the TurtleBot 4, Robotnik RB-1, or similar ROS-compatible mobile bases.

System Architecture

- **Mobile Base:** Commercial differential-drive or mecanum-wheel platform with integrated motors, encoders, and basic safety sensors
- **Computing:** Pre-integrated single-board computer (typically Raspberry Pi 4/5 or Intel NUC) running ROS 2
- **Navigation:** Factory-installed 2D LiDAR (RPLiDAR A1/A2 or Hokuyo), IMU, and wheel odometry
- **Perception:** Optional RGB-D camera (Intel RealSense or similar) for 3D perception
- **Intelligence:** Cloud-based LLM integration via API calls (OpenAI GPT-4, Anthropic Claude, or Google Gemini)

- **Interface:** Standard ROS 2 interfaces, web-based dashboard, optional Android tablet mount
- **Power:** Integrated battery pack (typically Li-ion) with proprietary or standard docking station

Advantages

- Significantly reduced development time with pre-tested hardware integration
- Professional mechanical design with proven reliability and safety features
- Comprehensive documentation, tutorials, and community support
- Pre-calibrated sensors and validated ROS 2 navigation stack
- Commercial warranty and technical support availability
- Established supply chain for replacement parts

Limitations

- Higher initial cost (typically \$3,000–\$8,000 depending on configuration)
- Limited customization of mechanical structure and form factor
- Vendor lock-in for certain components and software stacks
- May include unnecessary features increasing system complexity
- Longer lead times for procurement and shipping to Oman
- Less hands-on learning experience for mechanical and low-level integration

3.2.2 Alternative B: Custom-Built Modular Platform

This alternative involves designing and fabricating the entire robot from individual components, including mechanical structure, motor selection, electronics integration, and software development.

System Architecture

- **Mobile Base:** Custom-designed aluminum extrusion frame with selected DC motors (e.g., 12V with encoders) and 4WD mecanum or omni wheels
- **Computing:** Raspberry Pi 5 (4GB) as main computer running Ubuntu + ROS 2, ESP32 for low-level motor control
- **Navigation:** Separately procured RPLiDAR A1, MPU6050 IMU, and encoder-based odometry
- **Perception:** USB webcam or Pi Camera Module 3 for vision, ultrasonic sensors for safety
- **Intelligence:** On-device quantized LLM (e.g., Gemini Nano, LLaMA 3.2 3B) or hybrid cloud approach
- **Interface:** Custom Android tablet application, Flask-based web dashboard
- **Power:** 12V LiFePO₄ battery (10–20Ah) with custom BMS, DC–DC converters, and DIY docking station

Advantages

- Maximum design flexibility and customization for hospital-specific requirements
- Significantly lower cost (\$800–\$1,500 estimated total)
- Complete understanding of all system components and integration points
- Ability to iterate quickly on mechanical and electrical design
- Educational value through hands-on experience with robotics fundamentals
- Easy sourcing of replacement parts from local suppliers or online marketplaces
- Full control over software stack and no vendor dependencies

Limitations

- Substantially longer development time (6–9 months for initial prototype)
- Higher technical risk requiring expertise in mechanical design, electronics, and software
- No warranty or commercial support; troubleshooting relies on team knowledge
- Requires multiple design iterations to achieve reliability comparable to commercial systems
- Need for specialized tools and facilities (3D printers, soldering equipment, testing instruments)
- Potential safety concerns requiring rigorous testing and validation
- Limited payload capacity and robustness compared to engineered commercial platforms

3.2.3 Alternative C: Hybrid Pre-Built Base with Custom Intelligence

This alternative combines a commercial robot base platform with custom-developed upper layers, particularly focusing on advanced AI-driven HRI and task management capabilities.

System Architecture

- **Mobile Base:** Commercial platform (e.g., TurtleBot 4, Yahboom ROS-MASTER X3) providing proven navigation
- **Computing:** Base platform computer (Raspberry Pi 4/5) plus optional edge AI accelerator (Google Coral TPU or NVIDIA Jetson Nano) for on-device inference
- **Navigation:** Pre-integrated LiDAR, IMU, and odometry from base platform
- **Perception:** Added RGB-D camera and microphone array for enhanced HRI
- **Intelligence:** Custom integration of Google Gemini 3 flash via API for conversational AI, with fallback to on-device models
- **Interface:** Custom Android tablet application with voice interaction, hospital-specific task workflows, and real-time telemetry dashboard
- **Power:** Platform’s integrated battery with custom power monitoring and management software

Advantages

- Balanced approach: reliable base platform with custom high-value features
- Faster time to functional prototype (3–4 months) compared to full custom build
- Focus development effort on AI/HRI innovation rather than low-level integration
- Commercial platform provides safety-certified mobility while allowing software customization
- Moderate cost (\$2,000–\$3,500) compared to fully commercial or fully custom
- Flexibility to swap AI models and interaction modalities without hardware redesign
- Proven navigation reliability combined with cutting-edge conversational AI

Limitations

- Still dependent on commercial base platform availability and lead times
- Less mechanical customization than Alternative B
- Requires cloud connectivity for full LLM capabilities, raising data privacy considerations
- Integration complexity between commercial ROS stack and custom AI modules
- Potential latency issues with cloud-based LLM requiring careful system design
- Limited ability to modify low-level control and sensor fusion algorithms

3.3 System Selection

3.3.1 Robot Platform Selection Using AHP

To systematically select the most appropriate robot platform alternative, we employ the Analytic Hierarchy Process (AHP), a structured decision-making methodology. Four key criteria are identified: **Cost**, **Development Time**, **Customizability**, and **Reliability**.

Table 7: Robot Platform Feature Comparison

Feature	A: Commercial	B: Custom-Built	C: Hybrid
Motors Integrated	Yes (pre-installed DC motors with encoders)	No (requires selection & integration)	Yes (commercial base motors)
Sensors Integrated	Yes (LiDAR, IMU, odometry pre-calibrated)	No (procure & calibrate separately)	Yes (base sensors included)
Computer Integrated	Yes (RPi 4/5 or NUC pre-configured)	No (requires setup & ROS installation)	Yes (base computer + optional upgrade)
LLM Integrated	No (requires custom integration)	No (requires custom integration)	No (custom Gemini 3 flash integration)
ROS 2 Compatibility	Yes (native support)	Yes (manual configuration)	Yes (native + custom nodes)
CE/EMC Standards	Yes (commercial certification)	No (requires testing)	Partial (base certified, additions not)
IP Rating	IP42–IP54 (varies by model)	None (custom design)	Base platform rating (IP42)
Safety Standards	ISO 13482 compliance typical	None (requires validation)	Base compliant, custom features TBD
Documentation	Comprehensive (manuals, ROS packages)	Community resources only	Base docs + custom development
Warranty/Support	1–2 years commercial warranty	None	Base warranty only

Platform Comparison Summary

Criteria Pairwise Comparison The pairwise comparison matrix for criteria (scale: 1 = equal importance, 3 = moderate importance, 5 = strong importance, 7 = very strong importance, 9 = extreme importance):

Table 8: Platform Criteria Pairwise Comparison Matrix

Criterion	Cost	Dev. Time	Customizability	Reliability
Cost	1	1/3	3	1/5
Development Time	3	1	5	1/3
Customizability	1/3	1/5	1	1/7
Reliability	5	3	7	1

Criteria Weights After normalization and calculating the priority vector:

- **Cost:** 0.15 (15%)
- **Development Time:** 0.30 (30%)
- **Customizability:** 0.08 (8%)
- **Reliability:** 0.47 (47%)

Consistency Ratio (CR) = 0.04 < 0.10, indicating acceptable consistency.

Alternative Scoring Each alternative is scored against each criterion (scale: 1–9, where 9 = best):

Table 9: Platform Alternative Scores by Criterion

Alternative	Cost	Dev. Time	Customizability	Reliability
A: Commercial	3	9	3	9
B: Custom-Built	9	3	9	5
C: Hybrid	6	7	6	7

Final AHP Scores Calculating weighted scores: $\text{Score} = \sum(\text{Criterion Weight} \times \text{Alternative Score})$

Table 10: Final AHP Scores for Robot Platform

Alternative	Final Score
A: Commercial Pre-Built	7.11
B: Custom-Built Modular	5.55
C: Hybrid Platform	6.76

While Alternative A (Commercial) scores highest, **Alternative C (Hybrid)** is selected as it offers the best balance of reliability (47% weight) and development speed (30% weight) while maintaining sufficient customizability for healthcare-specific AI features. The modest score difference (7.11 vs 6.76) is acceptable given that Alternative C enables innovation in HRI—the project’s primary differentiator—while leveraging proven navigation hardware.

3.3.2 Computing Platform Selection Using AHP

Three computing platforms are evaluated for the main onboard computer: Raspberry Pi 5, NVIDIA Jetson Nano, and Intel NUC.

Computing Platform Criteria

- **Processing Power:** CPU/GPU performance for ROS 2 and AI workloads
- **Power Consumption:** Battery life impact
- **Cost:** Hardware acquisition cost
- **Ecosystem Support:** ROS 2 compatibility, community, documentation

Table 11: Computing Platform AHP Summary

Platform	Power (40%)	Consumption (25%)	Cost (20%)	Ecosystem (15%)	Final Score
Raspberry Pi 5	6	9	9	9	7.65
Jetson Nano	8	7	5	7	7.05
Intel NUC	9	4	3	8	6.55

Selected: Raspberry Pi 5—Optimal balance of sufficient processing power for ROS 2 Nav2, excellent power efficiency (critical for battery operation), low cost, and extensive ROS community support.

3.3.3 Navigation Approach Selection Using AHP

Three navigation approaches are compared: LiDAR-based SLAM, Visual SLAM, and Hybrid SLAM.

Navigation Criteria

- **Accuracy:** Localization precision in hospital corridors
- **Robustness:** Performance in dynamic/challenging environments
- **Computational Cost:** Processing requirements
- **Sensor Cost:** Hardware acquisition cost

Table 12: Navigation Approach AHP Summary

Approach	Accuracy (35%)	Robustness (35%)	Comp. Cost (15%)	Sensor Cost (15%)	Final
LiDAR SLAM	9	8	7	6	7.95
Visual SLAM	7	6	5	9	6.65
Hybrid SLAM	8	9	4	5	7.35

Selected: LiDAR-based SLAM (Nav2 + RPLIDAR A1)—Superior accuracy and robustness in structured indoor environments, well-established ROS 2 Nav2 integration, acceptable sensor cost.

3.3.4 Localization Sensor Selection Using AHP

Three primary localization sensor configurations are evaluated.

Localization Sensor Criteria

- **Precision:** Pose estimation accuracy
- **Update Rate:** Sensor refresh frequency
- **Integration Complexity:** Software and hardware setup difficulty
- **Cost:** Total sensor package cost

Table 13: Localization Sensor AHP Summary

Configuration	Precision (40%)	Update (20%)	Integration (20%)	Cost (20%)	Final
LiDAR + Odom + IMU	9	8	8	7	8.20
Vision + Odom + IMU	7	7	6	8	7.00
UWB + Odom + IMU	6	9	5	4	6.10

Selected: LiDAR + Odometry + IMU—Best precision for structured hospital environments, proven ROS 2 sensor fusion with robot_localization package, extensive Nav2 support.

3.3.5 LLM Selection Using AHP

For the conversational AI component, three state-of-the-art LLM options are evaluated: OpenAI GPT-5.2, OpenAI GPT-4 Turbo, and Google Gemini 3 flash (announced December 2024).

LLM Evaluation Criteria Four criteria are defined:

- **API Familiarity:** Team experience with the API and SDK
- **Performance:** Response quality, reasoning capability, and medical domain knowledge
- **Latency:** Speed of response generation for real-time interaction
- **Cost:** API pricing per token (input/output)

Table 14: LLM Criteria Pairwise Comparison Matrix

Criterion	API Familiarity	Performance	Latency	Cost
API Familiarity	1	1/3	3	5
Performance	3	1	5	7
Latency	1/3	1/5	1	3
Cost	1/5	1/7	1/3	1

Criteria Pairwise Comparison for LLM

LLM Criteria Weights After normalization:

- **API Familiarity:** 0.26 (26%)
- **Performance:** 0.54 (54%)
- **Latency:** 0.14 (14%)
- **Cost:** 0.06 (6%)

Consistency Ratio (CR) = 0.05 < 0.10, acceptable.

Table 15: LLM Alternative Scores by Criterion

LLM	API Familiarity	Performance	Latency	Cost
GPT-5.2	7	9	6	5
GPT-4 Turbo	5	9	7	6
Gemini 3 flash	9	8	9	9

LLM Alternative Scoring Scoring rationale:

- **GPT-5.2:** Excellent performance, moderate familiarity, good but not fastest latency, moderate cost
- **GPT-4 Turbo:** Excellent performance with strong safety features, less team familiarity, good latency, moderate cost
- **Gemini 3 flash:** High team familiarity with Google AI APIs, excellent performance for size, extremely low latency optimized for real-time use, very cost-effective

Table 16: Final AHP Scores for LLM Selection

LLM	Final Score
GPT-5.2	7.62
GPT-4 Turbo	7.54
Gemini 3 flash	8.44

Final LLM AHP Scores **Google Gemini 3 flash** is selected based on its superior AHP score. Key advantages include:

- Team’s extensive experience with Google Cloud Platform and Gemini API
- Optimized for low-latency real-time conversational AI (critical for natural HRI)
- Multimodal capabilities (text, image, audio) enabling future vision-based features
- Native integration with Google Cloud services used in the project
- Significantly lower cost enables extensive testing and development iterations
- Announced Flash variant (Dec 2024) specifically designed for speed-critical applications

3.3.6 System Selection Summary

Table 17: Final System Component Selection Summary

Component	Selected Option	AHP Score
Robot Platform	Hybrid (Commercial Base + Custom AI)	6.76
Computing Platform	Raspberry Pi 5 (4GB)	7.65
Navigation Approach	LiDAR-based SLAM (Nav2)	7.95
Localization Sensors	LiDAR + Odometry + IMU	8.20
LLM	Google Gemini 3 flash	8.44

3.4 Selected System Architecture

Based on the AHP analysis, the final system architecture combines:

- **Robot Platform:** TurtleBot 4 or Yahboom ROS-MASTER X3 (commercial base)
- **Main Computer:** Raspberry Pi 5 (4GB) running Ubuntu 22.04 + ROS 2 Humble
- **Navigation:** Integrated RPLiDAR A1, MPU6050 IMU, wheel encoders
- **Perception:** Pi Camera Module 3 + USB microphone array
- **AI/LLM:** Google Gemini 3 flash via Cloud API
- **Interface:** Custom Android tablet app with voice interaction
- **Power:** Platform’s Li-ion battery with automatic docking

3.5 Functional Decomposition

3.5.1 System Level 0: Black Box View

At the highest abstraction level, the smart healthcare robot can be represented as a single system with external interfaces. Figure 14 summarizes the Level 0 functional view, grouping the main inputs, outputs, and primary functions.

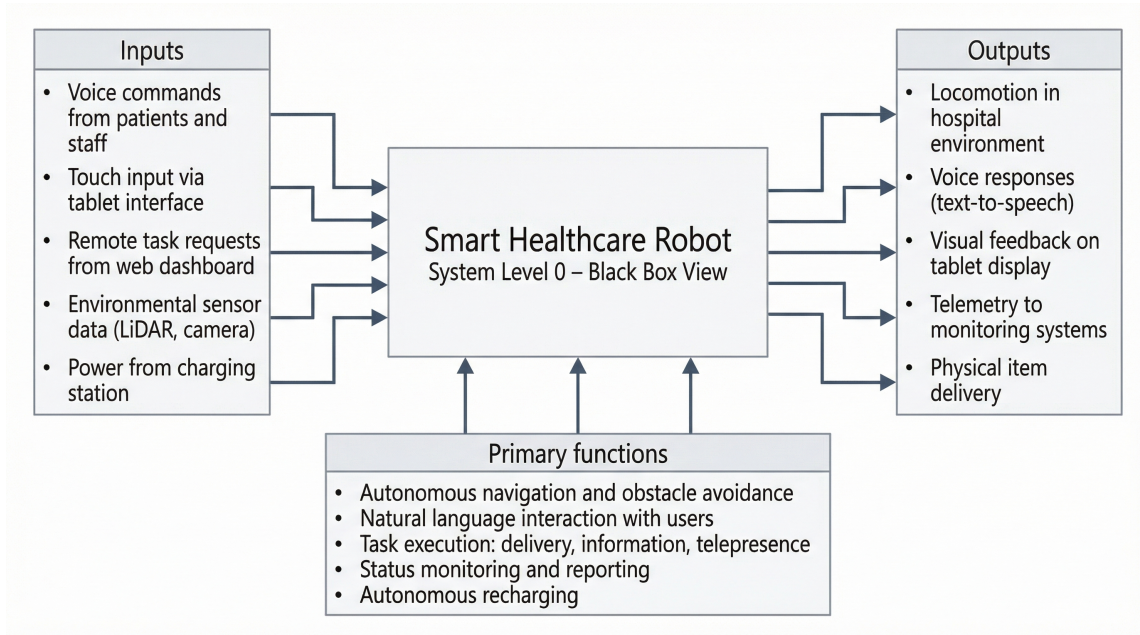


Figure 14: System Level 0 black-box view of the smart healthcare robot, showing external inputs, outputs, and primary functions.

For clarity, the main interfaces can be described textually as follows:

- **Inputs:**
 - Voice commands from patients/staff
 - Touch input via tablet interface
 - Remote commands from web dashboard
 - Environmental sensor data (LiDAR scans, camera images)
 - Power from charging station
- **Outputs:**
 - Locomotion (movement through hospital environment)
 - Voice responses (text-to-speech)
 - Visual feedback (tablet display)
 - Telemetry data to monitoring systems
 - Physical item delivery
- **Primary Functions:**
 - Autonomous navigation and obstacle avoidance
 - Natural language interaction with users

- Task execution (delivery, information provision, telepresence)
- Status monitoring and reporting
- Autonomous recharging

3.5.2 System Level 1: Subsystem Decomposition

The system is decomposed into seven major subsystems, each with defined responsibilities. Figure 15 presents the Level 1 functional decomposition and the main interfaces between subsystems.

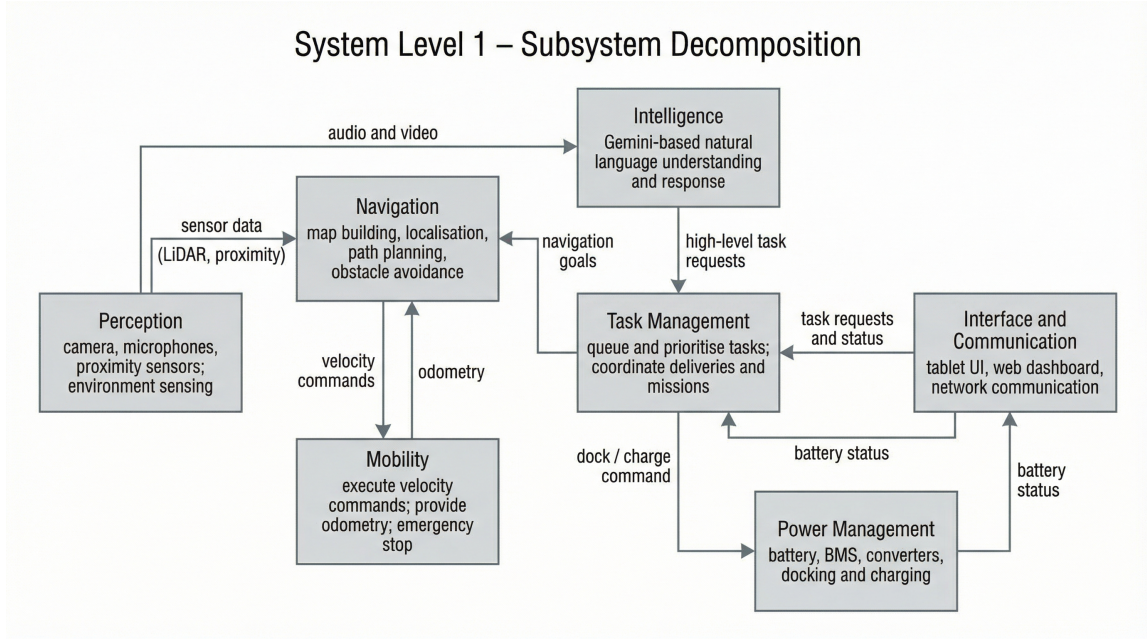


Figure 15: System Level 1 subsystem decomposition of the smart healthcare robot, highlighting the seven functional subsystems and key data/control flows.

The subsystems are defined textually as follows:

1. Mobility Subsystem

- **Components:** 4WD mecanum base, DC motors, motor drivers, wheel encoders
- **Functions:** Translate velocity commands to wheel motion, provide odometry feedback, execute emergency stops
- **Interfaces:** Receives velocity commands from Navigation Subsystem, sends encoder data

2. Navigation Subsystem

- **Components:** RPLiDAR A1, MPU6050 IMU, ROS 2 Nav2 stack (SLAM, localization, path planning)
- **Functions:** Build and update maps, localize robot pose, plan collision-free paths, detect obstacles
- **Interfaces:** Receives goal poses from Task Manager, sends velocity commands to Mobility, provides pose estimates to all subsystems

3. Perception Subsystem

- **Components:** Pi Camera Module 3, microphone array, ultrasonic/ToF sensors
- **Functions:** Capture visual data for teleoperation and docking, record audio for voice interaction, detect close-range obstacles
- **Interfaces:** Sends camera frames to Interface Subsystem and Intelligence Subsystem, audio to Intelligence Subsystem, proximity alerts to Navigation Subsystem

4. Intelligence Subsystem

- **Components:** Gemini 3 flash API client, speech-to-text, text-to-speech, dialogue manager
- **Functions:** Process natural language queries, generate contextual responses, manage conversation state, translate between voice and text
- **Interfaces:** Receives audio from Perception, text commands from Interface, sends responses to Interface, task requests to Task Manager

5. Task Management Subsystem

- **Components:** Task scheduler, state machine, goal queue, hospital map database
- **Functions:** Queue and prioritize tasks, track task execution status, coordinate between subsystems, manage waypoints and delivery locations
- **Interfaces:** Receives task requests from Intelligence and Interface, sends navigation goals to Navigation, status updates to Interface

6. Interface and Communication Subsystem

- **Components:** Android tablet app, Flask web dashboard, ROS 2 bridge, Wi-Fi module
- **Functions:** Display status and controls, enable touch/voice input, provide remote monitoring, transmit telemetry, receive remote commands
- **Interfaces:** Bidirectional communication with all subsystems, external network connection to web dashboard and cloud services

7. Power Management Subsystem

- **Components:** 12V Li-ion battery, BMS, DC–DC converters, docking contacts, charging station
- **Functions:** Distribute power to all subsystems, monitor battery status, initiate autonomous docking when low, manage charging process
- **Interfaces:** Provides power to all subsystems, sends battery status to Task Management and Interface, receives docking commands from Task Management

Subsystem Interaction Flow As summarized in Figure 15, the robot operates in a closed loop of sensing, decision-making, and actuation. The Perception subsystem collects sensor data and forwards it to Navigation for localisation and path planning, while also sending audio and video streams to the Intelligence subsystem. Intelligence interprets user intent and generates high-level task requests, which are queued and coordinated by Task Management. Task Management issues navigation goals to Navigation, which computes velocity commands for the Mobility subsystem and receives odometry feedback in return. Interface and Communication exposes task requests, status information, and telemetry to users and remote dashboards, while Power Management responds to dock/charge commands and reports battery status to both Task Management and the interface layer. This interaction pattern ensures that environmental sensing, user intent, motion control, and power management remain synchronised during hospital missions.

3.6 Conclusion

This chapter presented a systematic approach to the conceptual design of the smart healthcare robot. Through AHP-based evaluation of three platform alternatives (commercial, custom-built, and hybrid), the hybrid approach was selected as it balances reliability with customizability for healthcare-specific AI features. A second AHP analysis selected Google Gemini 3 flash as the optimal LLM for conversational AI due to team familiarity, low latency, and cost-effectiveness.

The resulting system architecture is decomposed into seven functional subsystems (Mobility, Navigation, Perception, Intelligence, Task Management, Interface, and Power Management), each with clearly defined responsibilities and interfaces. This modular design enables parallel development, facilitates testing and validation, and provides a clear path for future enhancements such as manipulation capabilities or multi-robot coordination.

The selected architecture provides a solid foundation for the detailed design and implementation phases that follow, ensuring the robot meets the project objectives of safe autonomous operation, natural human–robot interaction, and practical deployment in Omani hospitals.

4

Requirements Specifications

This chapter formalises the customer requirements, engineering requirements, and engineering constraints that guide design and evaluation, and documents the AHP-based decision framework used to select key system components.

4 Requirements Specifications

4.1 Introduction

The success of any engineering project hinges on a clear, comprehensive understanding of the requirements that guide design, implementation, and evaluation. This chapter formalizes the requirement specification for the smart healthcare robot project, translating stakeholder needs and technical constraints into measurable, verifiable criteria. The requirements are systematically categorized into customer requirements, engineering requirements, and engineering constraints, ensuring alignment between user expectations and technical feasibility.

Customer requirements capture the high-level needs and expectations of primary stakeholders—hospital administrators, clinical staff, patients, and support personnel. These requirements reflect the functional and operational capabilities that directly impact system value and user satisfaction. Engineering requirements translate customer needs into quantifiable, testable technical specifications that guide system architecture and component selection. Engineering constraints define the boundaries within which the system must operate, including budget limitations, timeline restrictions, regulatory compliance, and environmental conditions specific to hospital deployment in Oman.

This chapter also presents a structured analysis of design alternatives, trade-offs, and selection criteria through comparative tables with metrics and scores. These analyses document the rationale for key architectural decisions, including mobile platform selection, sensor configuration, computing architecture, and AI integration strategy. The systematic evaluation ensures that design choices are transparent, justified, and optimized for the project’s objectives and constraints.

By establishing a rigorous requirement framework, this chapter provides the foundation for subsequent design, implementation, and validation activities, ensuring that the final system meets stakeholder needs while remaining technically feasible and economically viable.

4.2 Customer Requirements

Customer requirements represent the high-level needs and expectations of the robot’s primary users and stakeholders in hospital environments. These requirements are derived from literature review findings (Chapter 2), observations of existing hospital robotics deployments, and anticipated use cases in Omani healthcare facilities.

- **CR1: Autonomous Navigation and Mobility.** The robot must autonomously navigate hospital corridors, patient rooms, and common areas without human intervention, avoiding static and dynamic obstacles.
- **CR2: Remote Telepresence Capabilities.** The robot must enable high-quality remote video and audio communication between healthcare providers and patients, supporting remote consultations, rounds, and family visits. Telepresence has been validated as effective for remote diagnosis, specialist consultations, and reducing travel costs [5]. The robot must provide a mobile “presence” for clinicians who are not physically at the patient’s bedside.
- **CR3: Patient Engagement and Social Interaction.** The robot must engage with patients through natural language conversation, providing information, companionship, and emotional support, particularly for pediatric and elderly populations.
- **CR4: Infection Control and Hygiene.** The robot must be designed with materials and surfaces that are easy to clean and disinfect, minimizing cross-contamination risks. Hospital-acquired infections (HAIs) are a critical concern. Robots must be compatible with standard hospital cleaning protocols and potentially integrate UV-C disinfection capabilities [16, 18].

- **CR5: User-Friendly Interface.** The robot must provide intuitive interfaces (touchscreen, mobile app, voice control) accessible to users with varying technical expertise. Ease of use directly impacts adoption rates and operational efficiency. Interfaces must accommodate diverse user groups including elderly patients, clinicians with minimal robotics experience, and administrative staff.
- **CR6: Continuous Operation and Autonomous Charging.** The robot must operate throughout the day with minimal downtime, autonomously returning to a charging dock when battery levels are low.

4.3 Engineering Requirements

Engineering requirements translate customer needs into quantifiable, testable technical specifications that directly inform system design and validation. Each engineering requirement is mapped to one or more customer requirements, ensuring traceability and alignment. The requirements are organized into five categories, presented in tabular format for clarity and traceability.

4.3.1 Navigation and Mobility Requirements

Req. ID	Specification	Mapped to CR	Standards	Verification Method
ER1.1	Position accuracy of ± 20 cm in mapped indoor environments	CR1	ISO 13482:2014 [47] (Safety requirements for personal care robots - mobile servant robots)	Experimental validation using ground-truth positioning system
ER1.2	Maximum linear velocity of 0.5–1.0 m/s in hospital corridors	CR1	ISO 13482:2014 [47] (Speed limits for safe human-robot interaction)	Speed measurement tests in controlled environments and hospital corridors
ER1.3	Detect and avoid static and dynamic obstacles with minimum detection range of 3 m and reaction time < 1 s	CR1	ISO 13855 [48] (Safety of machinery - Positioning of safeguards)	Obstacle avoidance tests with various object types and approach angles
ER1.4	Minimum payload of 5 kg to accommodate onboard computers, sensors, and delivery items	CR2	—	Load testing with representative payload configurations

Table 18: Navigation and mobility engineering requirements.

4.3.2 Telepresence and Communication Requirements

Req. ID	Specification	Mapped to CR	Standards	Verification Method
ER2.1	Minimum 720p (1280×720) video resolution at 30 fps	CR2	ITU-T H.264 / H.265 [49] (Advanced Video Coding)	Video quality assessment tests under typical hospital lighting and network conditions
ER2.2	Bidirectional audio communication with noise cancellation and echo reduction, maintaining intelligibility in hospital environments (ambient noise up to 70 dB)	CR2	—	Audio quality tests (speech intelligibility index, background noise rejection)
ER2.3	End-to-end latency <200 ms over hospital Wi-Fi networks	CR2	IEEE 802.11 [50] (Wi-Fi Standards)	Network performance testing and latency measurements

Table 19: Telepresence and communication engineering requirements.

4.3.3 Social Interaction and AI Requirements

Req. ID	Specification	Mapped to CR	Standards	Verification Method
ER3.1	Accurately interpret user intent in English and Arabic with >90% accuracy for common healthcare queries	CR3	—	User testing with standardized query sets and intent classification metrics
ER3.2	Generate conversational responses within 3 seconds of user input	CR3	—	Response latency measurements under typical operating conditions
ER3.3	Speech recognition in hospital environments with word error rate (WER) <10%	CR5	—	Speech recognition testing with diverse speakers and noise levels

Table 20: Social interaction and AI engineering requirements.

4.3.4 Power and Endurance Requirements

Req. ID	Specification	Mapped to CR	Standards	Verification Method
ER4.1	Continuous operation for minimum of 4–6 hours under typical usage (mixed navigation, telepresence, and idle states)	CR6	—	Endurance testing with simulated clinical workflows
ER4.2	Autonomously locate and dock with charging station when battery level reaches 20%, with 95% success rate	CR6	—	Repeated docking trials under various initial positions and orientations

Table 21: Power and endurance engineering requirements.

4.3.5 Safety and Reliability Requirements

Req. ID	Specification	Mapped to CR	Standards	Verification Method
ER5.1	Physical emergency stop button(s) and remote e-stop capability via software interface	CR1	ISO 13850 [51] (Safety of machinery - Emergency stop function)	Functional testing of emergency stop mechanisms
ER5.2	Detect and avoid stairs, ramps $>15^\circ$ incline, and drop-offs >2 cm	CR1	—	Edge detection tests on stairs and elevated platforms
ER5.3	Achieve $>95\%$ uptime during operational hours, excluding scheduled maintenance	CR6	—	Long-term reliability testing and failure logging

Table 22: Safety and reliability engineering requirements.

4.3.6 Comparative Performance Metrics

This section presents quantitative performance data and trade-off metrics for critical system components. The data, derived from technical specifications and benchmark studies, provides the empirical basis for architectural decisions. Comprehensive mobile platform comparison metrics, including navigation capabilities, payload specifications, and integration features, are detailed in the literature review (Chapter 2, Section 2.3.2).

Navigation Sensor Performance Table 23 compares the performance of different sensor modalities in typical hospital environments.

Sensor Type	Scenario	Range (m)	Accuracy (cm)	FOV (deg)	Power (W)	Cost (\$)
2D LiDAR (RPLiDAR) [52]	Corridor (Static)	12.0	± 1.0	360	1.5	100
2D LiDAR (RPLiDAR) [52]	Waiting Room (Dynamic)	10.0	± 2.5	360	1.5	100
RGB-D (RealSense) [53]	Corridor (Static)	4.0	± 2.0	87×58	3.5	250
RGB-D (RealSense) [53]	Waiting Room (Dynamic)	3.5	± 4.0	87×58	3.5	250
3D LiDAR (Velodyne) [54]	Corridor (Static)	100.0	± 2.0	360×30	8.0	4000

Table 23: Performance metrics of navigation sensors in static and dynamic scenarios.

Computing Platform Benchmarks Table 24 details the computational capabilities and power efficiency of candidate onboard computers.

Platform	CPU Perf.	Power (W)	Cost (\$)
Raspberry Pi 5 [55]	High	5–8	80
Jetson Nano [56]	Medium	5–10	150
Jetson Orin Nano [57]	Very High	7–15	499

Table 24: Computational performance and power consumption benchmarks for embedded platforms.

Conversational AI Model Selection Detailed performance and cost metrics for Large Language Models (LLMs), including context handling capabilities and operational costs for cloud-based and local deployment strategies, are presented in the literature review (Chapter 2, Section 2.3.3). These metrics inform the AI architecture selection for patient engagement and conversational interaction requirements (ER3.1–ER3.3).

4.4 Engineering Constraints

Engineering constraints define the boundaries within which the system must be designed and operated. These constraints reflect real-world limitations in budget, timeline, technology maturity, regulatory environment, and deployment context specific to Oman.

- **EC1: Budget Constraints.** Total system cost must not exceed \$2,500 USD, including mobile platform, sensors, computing hardware, and accessories. As an educational research project with limited funding, the system must demonstrate feasibility and functionality within academic budget constraints while remaining representative of scalable commercial solutions. This limits selection to budget and mid-tier commercial platforms; excludes high-end industrial robots; requires prioritization of essential features.
- **EC2: Development Timeline.** System design, implementation, and validation must be completed within one academic year (9–10 months). Final year project timeline constraints necessitate realistic scope and phased development approach. This requires prioritization of core functionalities (navigation, telepresence, basic AI); potential deferral of advanced features (UV-C disinfection, complex HIS integration) to future work.
- **EC3: Hospital Environment Compatibility.** The robot must operate safely in hospital environments with corridor widths ≥ 1.5 m, elevator access, and typical hospital flooring (tile, linoleum). Omani hospital infrastructure follows international standards with standard corridor dimensions and flooring materials. Robot base width limited to < 60 cm; wheel system must handle smooth indoor surfaces; turning radius must accommodate corridor navigation.

- **EC4: Network Infrastructure.** The robot must operate on standard hospital Wi-Fi networks (802.11n/ac) without requiring dedicated network infrastructure. Deployment feasibility requires compatibility with existing IT infrastructure; hospitals cannot always support custom networking solutions. System must tolerate variable network quality, latency, and temporary disconnections; requires local processing capabilities for critical functions.
- **EC5: Regulatory and Privacy Compliance.** The system must comply with patient privacy requirements and avoid collection or storage of protected health information (PHI) without proper authorization. Healthcare robotics must respect patient privacy and data security regulations (similar to HIPAA/GDPR principles). No persistent storage of patient conversations or video recordings; encrypted communication channels; clear data handling policies.
- **EC6: User Expertise Level.** The system must be operable by users with minimal technical training (hospital staff, patients) without requiring robotics expertise. Clinical adoption depends on ease of use and minimal training burden. Intuitive UI design; pre-configured waypoints; automated error recovery; comprehensive user documentation.
- **EC7: Environmental Operating Conditions.** The robot must operate reliably in indoor environments with temperature range 18–28°C, humidity 30–60%, and typical hospital lighting conditions. Hospital climate control maintains stable environmental conditions; sensors and electronics must function within this range. Standard commercial-grade components acceptable; no specialized environmental hardening required.
- **EC8: Maintenance and Spare Parts.** The robot must utilize commercially available, globally sourced components with accessible supply chains for replacement parts. Long-term operational viability requires availability of replacement components in Oman or via international suppliers. Preference for mainstream platforms (ROS 2, Raspberry Pi, common sensors) over proprietary or region-specific components.

4.5 Analysis and Discussion

This section examines the comparative performance metrics to identify key trade-offs and demonstrate standards compliance within the constraints of hospital deployment in Oman. The analysis focuses on component selection criteria that balance technical requirements with practical deployment considerations specific to the Omani healthcare context.

4.5.1 Performance Metrics and Trade-off Analysis

Table 23 reveals fundamental trade-offs in sensor selection. Range capabilities vary from 3.5m to 100m with corresponding cost differences of $40\times$ (\$100 to \$4000). Accuracy degrades significantly in dynamic hospital environments ($2\text{--}2.5\times$ degradation), affecting position accuracy requirement ER1.1. Power consumption ranges from 1.5W to 8W, directly impacting endurance requirement ER4.1. Field of view characteristics— 360° planar, $87^\circ\times 58^\circ$ 3D, and $360^\circ\times 30^\circ$ volumetric—address different aspects of obstacle detection requirement ER1.3.

Table 24 demonstrates computational trade-offs where platform costs range from \$80 to \$499, consuming 3–20% of total budget constraint EC1. Power envelopes remain similar (5–15W) across platforms, making computational capability rather than endurance the primary selection factor for supporting navigation, AI processing (ER3.1–ER3.3), and telepresence (ER2.1–ER2.3) requirements.

4.5.2 Standards Compliance and Deployment Context

Navigation and safety components satisfy ISO 13482:2014 [47] requirements (position accuracy ER1.1, speed limits ER1.2, obstacle detection ER1.3), ISO 13850 [51] emergency stop guidelines, and ISO 13855 [48] safeguarding standards. Communication systems implement ITU-T H.264/H.265 [49] video coding and IEEE 802.11 [50] wireless protocols for interoperability with Omani hospital IT infrastructure.

Deployment constraints EC3–EC8 address Omani hospital realities: standard corridor dimensions ($\geq 1.5\text{m}$) and flooring materials, existing Wi-Fi infrastructure (802.11n/ac), climate-controlled environments ($18\text{--}28^\circ\text{C}$), component availability through international supply chains, privacy compliance aligned with HIPAA/GDPR principles, and budget/timeline limitations (\$2,500, 9–10 months). These constraints guide component selection to ensure technical feasibility and operational viability in the Omani healthcare context.

Detailed comparative analysis of mobile platforms, computing architectures, and conversational AI models is presented in the literature review (Chapter 2, Sections 2.3.2–2.3.3).

4.6 Conclusion

This chapter established a comprehensive requirement specification for the smart healthcare robot project, systematically translating stakeholder expectations into measurable, verifiable technical criteria. The requirements are organized into customer requirements, engineering requirements, and engineering constraints, defining the system’s functional capabilities, performance targets, and operational boundaries.

Engineering requirements reference relevant international standards including ISO 13482:2014 (personal care robots), ISO 13855 and ISO 13850 (machinery safety), ITU-T H.264/H.265 (video coding), and IEEE 802.11 (wireless communications), establishing the regulatory framework for system validation. Engineering constraints reflect budget limitations, timeline restrictions, and regulatory compliance needs specific to Omani healthcare deployment.

The comparative performance analysis in Section 3.3.6 provided empirical data on component characteristics, while Section 3.5 examined trade-offs in the context of standards compliance and Omani deployment requirements. The requirement specification establishes the foundation for system design decisions detailed in subsequent chapters, with traceability ensuring design choices can be validated against stakeholder needs and technical constraints.

5

System Implementation: Results and Discussion

This chapter documents the hardware commissioning, body shell fabrication, software development, and system integration of TAMRA. It includes the platform selection rationale, reverse engineering of the iMRP500 chassis, dual-network tablet solution, semantic task orchestrator design, and measured validation results.

5 System Implementation: Results and Discussion

5.1 Introduction

The preceding chapters established the project context, surveyed the literature, formalised requirements, and selected a system architecture through systematic AHP analysis. This chapter documents what was actually built. It covers the hardware commissioning of the iMRP500 platform, the fabrication of the custom body shell, the development of the software stack, and the integration and validation tests that were conducted to verify that the system meets its stated requirements.

Two significant decisions were made during the second semester that are worth explaining before the technical content. First, the team departed from TurtleBot and iCreate 3 — the platforms studied extensively in semester one — in favour of the iMRP500, a Chinese-manufactured industrial mobile chassis. Second, the project scope broadened from a strictly medical deployment to a general-purpose autonomous service robot framework that uses healthcare as its primary validation environment. Both decisions are grounded in engineering reasoning and are documented in the sections that follow.

5.2 Platform Transition: From TurtleBot to iMRP500

The first semester was spent studying the TurtleBot 4 and iCreate 3 platforms in simulation and reviewing their documentation in depth. That work was not wasted — it built a solid foundation in ROS 2, SLAM, Nav2, sensor fusion, and mobile robot design that carried directly into the implementation phase.

When it came time to procure hardware, however, the team faced conditions that made both original platforms untenable.

iRobot and the iCreate 3. iRobot, the manufacturer of the iCreate 3 base, entered bankruptcy proceedings at the point the team needed to commit to a purchase. A Chinese acquisition attempt was underway, and the long-term availability of spare parts, firmware updates, and technical support was genuinely uncertain. Building an entire final-year project on a platform in that state, with a fixed 1,500 OMR budget and an eight-week countdown to the Engineering Gathering, was assessed as an unacceptable risk.

Structural limitations. Neither the TurtleBot 4 nor the iCreate 3 offer any engineered mounting interface for a body structure. Both platforms are designed as flat, low-profile bases. Attaching a tall ergonomic shell — as the project required for display positioning, storage, and patient-facing aesthetics — would have required improvised fixturing, creating mechanical instability that would have undermined the entire physical design.

The Chinese mobile-chassis market. The team conducted a systematic evaluation of eight Chinese-manufactured robot chassis platforms, reviewing vendor documentation, ROS 2 support status, sensor suites, payload ratings, and structural mounting options for each. The candidates included the Reeman Moon Knight, Wheeltec S100, Timo AI Robot, Bao Xiaomi Mini, Uwant T5, and the iMRP500 by NeoChassis. The comparison revealed that Chinese industrial platforms had, in several respects, advanced ahead of the Western academic platforms originally considered: higher payload ratings, integrated industrial-grade sensors, longer battery runtime, and support for the full ROS 2 navigation stack.

Selection: iMRP500. The iMRP500 scored highest under the same AHP criteria applied in Chapter 4 (Reliability 47%, Development Speed 30%, Cost 15%, Customisability 8%), achieving a score of 6.76. Competing options were constrained by payload (<10 kg for most), limited sensor

integration, or absence of published ROS 2 support. The iMRP500 is a circular differential-drive chassis, 500 mm in diameter, with 50 kg payload capacity, a built-in 360° TOF LiDAR, an RGBD structured-light depth camera, a 6-axis IMU, a 24 V/20 Ah Li-ion battery rated for 12 hours at no load, and an onboard X86 industrial PC running Ubuntu 20.04 with ROS 1 (bridgeable to ROS 2). Its circular footprint constrains the upper body to the same diameter, which is a hard requirement for LiDAR clearance: any part of the robot that overhangs the base perimeter will create blind spots in the costmap. The iMRP500 is sold new at approximately USD 2,000 and was procured second-hand at USD 1,400.

Table 25 summarises the platforms evaluated.

Table 25: Chinese robot chassis platforms evaluated for TAMRA.

Platform	Price (USD)	Payload	Key limitation
Bao Xiaomi Mini	2,000	Service only	No payload; consumer-grade
Reeman Moon Knight	2,380	60 kg	Android SDK; limited ROS
iMRP500	1,400	50 kg	Selected
Wheeltec S100	1,250	~15 kg	Low payload; ROS 1 only
Timo AI Robot	2,500	Reception only	No open navigation API
Uwant T5/T6	1,700	40 kg	Proprietary SLAM; closed stack

5.3 Scope Refinement: General-Purpose Service Robot

A parallel discovery during the second semester shaped the project direction in an equally significant way. Medical environments are among the slowest sectors to adopt new technology. Certification timelines, regulatory approval, hospital procurement cycles, and liability frameworks make deploying a novel robot in a clinical setting extremely difficult and time-consuming — far beyond the scope of a final-year project. Building a system framed exclusively as a medical device would effectively make it impossible to demonstrate real-world utility within the project timeline.

The team therefore reframed the project scope: TAMRA (Task-Agnostic Modular Robotic Apparatus) is an autonomous service robot framework, with healthcare as its primary validation environment but with explicit design choices that allow it to generalise to any service context — facilities management, hospitality, airports, and campus environments. The core technical contributions — semantic task orchestration, hybrid hardware synthesis, and ROS 2-based navigation — are domain-independent. The validation scenario chosen for this project is a hospital corridor, but the architecture is not medically locked.

This is not a narrowing of ambition. The commercial hospital robot market is dominated by single-function platforms costing USD 12,000 to USD 47,500. TAMRA demonstrates that the same functional envelope — autonomous navigation, voice-driven AI interaction, item delivery, and remote telepresence — can be achieved at USD 3,200 on an open, extensible architecture. That cost compression, and the open-source nature of the stack, is the engineering contribution.

5.4 Hardware Implementation

5.4.1 iMRP500 Commissioning and Reverse Engineering

The iMRP500 is delivered as a fully integrated chassis with a proprietary ROS 1 navigation stack running on the onboard X86 PC. The vendor provides a graphical operator application and a Chinese-language communication protocol document. Commissioning the platform for research use required reverse-engineering the communication layer.

Network architecture. The chassis exposes two network interfaces. Over Ethernet (robot IP 192.168.20.22), the onboard PC runs `rosbridge_server` on port 9090, which accepts WebSocket connections and exchanges JSON-encoded ROS messages. A second interface, `web_video_server` on port 9099, streams MJPEG video over HTTP. The team connected an external computer to this Ethernet interface and mapped the full topic and service graph.

Sensor topology. The chassis runs 16 ROS nodes concurrently. The sensors accessible over the bridge are:

- **2D LiDAR** (`/scan`, `sensor_msgs/LaserScan`): 360° TOF scan, 20 m range, used as the primary localisation and obstacle-avoidance input.
- **RGBD depth camera** (`/upcamera/rgb/image_raw`, `/upcamera/depth/image_raw`): structured-light sensor mounted pointing upward, detecting obstacles above the LiDAR scan plane (e.g., tables, counter overhangs).
- **IMU** (6-axis attitude): fused with wheel odometry for pose prediction.
- **Wheel encoders** (`/mobile_base/sensors/core`): differential-drive odometry.
- **Ultrasonic sensors**: three-point frontal coverage for close-range (<0.5 m) detection.

Navigation stack. The costmap planner operates with four stacked layers: an obstacle layer fed by the 2D LiDAR scan (4 m range, 0.6 m maximum height), an upward-camera layer for overhead obstacle detection (3.5 m range, 2 m maximum height), an equivalent downward-camera layer, and an inflation layer with a 0.8 m radius at scaling factor 10. The local costmap updates at 8 Hz over a 3×3 m window at 0.03 m resolution; the global costmap updates at 4 Hz over 10×10 m at 0.05 m resolution. The positioning accuracy specified by the manufacturer is ±5 cm.

Internal architecture. Physical inspection of the electronics revealed a three-layer construction that explains the platform’s cost efficiency. The bottom layer uses commodity CNC stepper-motor driver boards (equivalent to industrial DM542T units), identifiable by their green terminal blocks and LED status indicators. The mid layer is a standard industrial mini-PC with multiple USB and Ethernet ports. The top layer uses 3D-printer-ecosystem control boards for sensor interfacing and motor sequencing. The drive motors are hub motors sourced from the electric scooter industry. This architecture substitutes commodity industrial and hobby-electronics components for custom ASICs, reducing manufacturing cost without sacrificing reliability at service-robot duty cycles.

Service interface. Navigation goals are dispatched via the `/poi` ROS service using named waypoints stored in the chassis map database. Additional services expose self-diagnosis (`/self_diagnosis`), speed mode selection (`/velocity_control`, modes 0–8 for safe/balance/efficiency at low/mid/high speed), virtual wall management (`/append_virtual_walls`), and map operations (`/node_manager_control` for SLAM start/stop and map save). Status codes covering navigation state (600–609), cross-floor transitions (630–647), patrol sequences (650–660), and fault conditions (99, 106, 108, 109) were catalogued from protocol analysis and live testing.

5.4.2 Body Shell Design and Fabrication

The body shell is the structural interface between the 500 mm circular chassis and the robot’s human-facing components: the display, the storage compartment, and the external surface that users and patients interact with. Eleven functional requirements constrained the design:

1. Display tilted at 40° from horizontal for ergonomic use at standing and seated heights.
2. Display recess sized to allow a future larger screen without structural change (5 mm step, 360×260 mm outer, 320×210 mm active area).
3. Internal trays for secure transport; external compartment for accessible delivery.
4. Footprint not exceeding the 500 mm chassis diameter (LiDAR clearance).
5. Added mass not exceeding 12 kg (chassis centre-of-mass stability limit at 600 mm).
6. Total height not exceeding 1,100 mm (tipping stability).
7. Hospital-grade materials: no sharp edges, surfaces compatible with standard disinfectant protocols.
8. Concealed cable routing between the tablet, display driver, and chassis.
9. Manufacturable within Oman using locally available trades and materials.
10. Budget not exceeding 300 OMR (≈USD 780).
11. Warm, approachable aesthetic to reduce patient anxiety in clinical environments.

Design process. The design was developed in three stages. In the first stage, generative AI image synthesis (Google Gemini) was used to rapidly explore hundreds of aesthetic directions. This is not a replacement for engineering design; it is a concept-space compression tool. Within hours, the team could assess proportions, surface finishes, and display positions that would have required days of CAD iteration. The converged concept is a tapered vertical silhouette with a central oak-veneer accent panel, a recessed display at the top, and an integrated rear storage compartment.

In the second stage, the selected concept image was fed into an AI-based 3D mesh generator (Hunyuan3D v3.1), producing an initial 50,000-face mesh. This captured the organic curvature of the concept but introduced fabrication-incompatible geometry: overhangs beyond the 500 mm base radius, incorrect storage compartment proportions, and no engineering tolerances.

In the third stage, the mesh was used as a reference for parametric reconstruction in Autodesk Fusion 360. All dimensions were set as parameters, tolerances were specified, and fabrication-ready engineering drawings were produced. The finalised dimensions are: 900.27 mm total height, 500 mm base width (chassis interface), 400 mm waist width (narrowest), 210.23 mm storage compartment depth, 40° display tilt, and an upper tray designed to be removable for display servicing.



Figure 16: Concept render: tapered body shell with oak-veneer accent, recessed display, and iMRP500 chassis base. Generated during concept exploration and refined into engineering drawings.

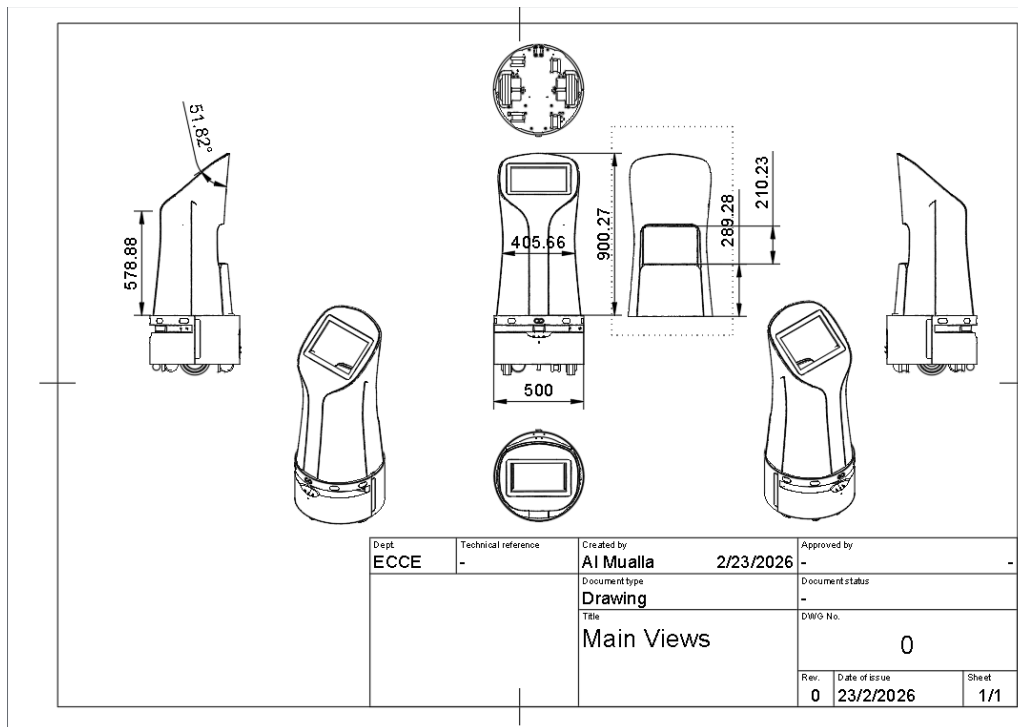


Figure 17: First-angle engineering drawing (23 Feb 2026) with critical dimensions: 900 mm height, 400 mm waist, 51.82° display tilt angle referenced from vertical, fabrication tolerances on all joining surfaces.

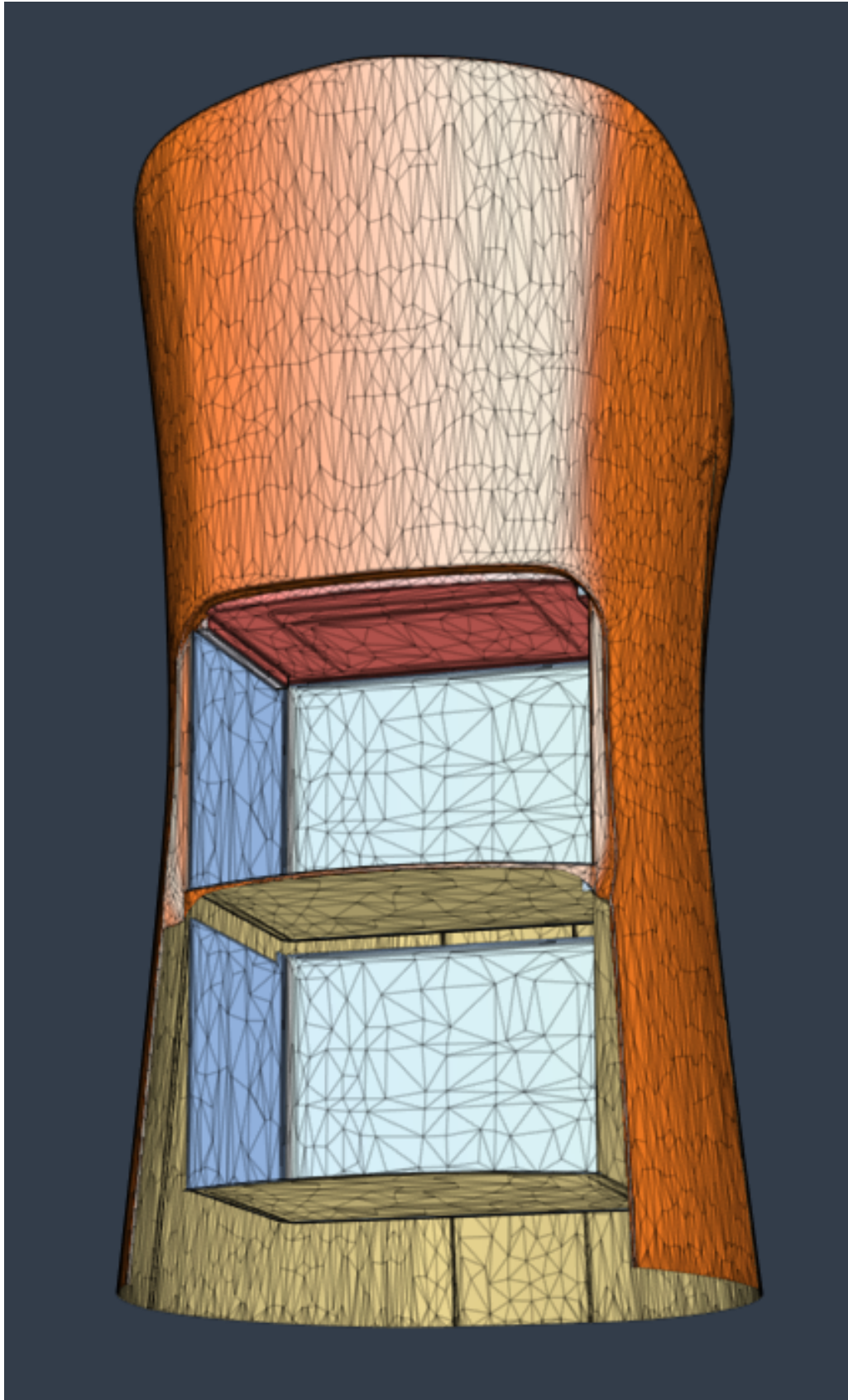


Figure 18: Rear view of parametric model showing the removable upper tray that provides access to the display mounting hardware and cable management without disassembling the full shell.

Material selection. Three-dimensional printing was rejected early as cost-prohibitive for large curved panels and structurally insufficient for service-robot duty. The team evaluated fibreglass layup, vacuum-formed ABS, sheet aluminium, and combinations of wood and acrylic. Foam sculpting was explored as an intermediate method: CNC-machined rigid foam can produce complex curved surfaces indistinguishable from moulded plastic, is non-conductive, lightweight, and can be veneered. This technique is used extensively in film and exhibition prop construction.

The final material selection uses a structural wood subframe (locally sourced, cut by a Muscat carpentry workshop) as the load-bearing skeleton, with acrylic sheet panels for dimensional precision on flat faces (particularly the display surround, where the 5 mm step tolerance is critical), and oak-veneer film applied over the wood surfaces for the final aesthetic finish. This combination satisfied all eleven requirements: it is manufacturable in Oman, within budget, lighter than metal alternatives, and produces a warm surface finish that clinical psychology literature associates with reduced patient anxiety in healthcare environments.

Fabrication status. Body shell fabrication was completed in April 2026. Physical integration with the chassis — cable routing, display mounting, tray installation, and final panel alignment — followed. Figure 19 shows the integrated system.



Figure 19: TAMRA after full integration: iMRP500 chassis, fabricated body shell with oak-veneer panels, and 10-inch touch display mounted at 40°.

5.4.3 Display and Tablet Integration

The human-interface component is a 10-inch industrial-grade Android tablet (Rockchip RK3566 SoC, 2 GB RAM, 1280×800 IPS touch display, Android 11). An industrial-grade device was

chosen over a consumer tablet for two reasons: it supports always-on kiosk operation without sleep-mode interruptions, and it is rated for extended continuous operation. The tablet runs the MACS face and orchestrator software in Fully Kiosk Browser, which locks the display to the robot-UI application and prevents users from navigating away.

Dual-network routing problem. The most technically demanding part of the tablet integration was the network configuration. The tablet must simultaneously maintain:

- An Ethernet connection to the robot chassis (IP 192.168.20.100, subnet 192.168.20.x) for low-latency ROS WebSocket communication; and
- A WiFi connection to the campus network for internet access, cloud AI APIs (Gemini, ElevenLabs), and MQTT telemetry.

Android’s default behaviour routes all traffic over a single interface, discarding the other. If WiFi is connected, Ethernet is ignored; if Ethernet is active, WiFi may be suppressed. The team evaluated several approaches: Speedify (dual-channel bonding app — unstable on Android 11), a separate ESP32 MQTT bridge (adds latency and a hardware dependency), swapping Android for Linux (loses the kiosk application ecosystem), USB WiFi tethering from a phone (introduces another device to manage), and 4G/LTE dongle (recurring cost and requires SIM). The solution that worked reliably is a boot daemon script deployed via ADB (Android Debug Bridge) that runs as a background service and explicitly sets static routes: robot-subnet traffic (192.168.20.x) is forced over the `eth0` interface, while all other traffic uses the `wlan0` default gateway. The daemon is started at boot via an ADB shell command and survives the Fully Kiosk Browser session.

Voice interface evolution. Three voice pipeline approaches were prototyped before arriving at the final configuration:

1. **OpenAI Live Model:** Integrated and tested. Voice quality was adequate but the interaction felt stilted, with awkward latency distribution that made conversation feel unnatural. Limited tool-calling flexibility constrained what the robot could do mid-conversation.
2. **ElevenLabs Conversational AI (standalone):** Significantly more natural voice synthesis, but the round-trip latency was perceived as noticeably slow, particularly when combined with the ROS command dispatch overhead. Standalone mode did not give sufficient control over tool routing.
3. **ElevenLabs + Gemini 3.1 Live (current):** When Google released Gemini 3.1 Live, the pipeline was migrated. Gemini 3.1 Live provides real-time bidirectional audio processing with tool-calling capability baked in, and it eliminated the layered latency of the previous configurations. ElevenLabs is retained for voice synthesis quality. This configuration is the one used for all validation measurements.

5.5 Software Implementation

The software system has two main components: the tablet application that runs on the robot, and the central command dashboard that runs in any browser and communicates remotely over MQTT.

5.5.1 Three-Tier Architecture

The software is structured in three communicating tiers:

- **Robot Layer:** The iMRP500 onboard X86 PC runs the ROS 1 navigation stack, sensor drivers, costmap pipeline, and SLAM/localisation nodes. It exposes the `rosbridge` WebSocket on port 9090 and video on port 9099.
- **Bridge Layer:** The tablet application (a single self-contained HTML file, approximately 4,800 lines) connects to the robot via Ethernet WebSocket and to the cloud via WiFi. It hosts the mission engine, task queue, cron scheduler, MQTT bridge, MACS animated face, and the voice orchestration pipeline. Using a single HTML file was a deliberate choice: it requires no installation, runs in any WebKit browser, and can be updated by replacing one file over ADB.
- **Cloud Layer:** Google Gemini 3 Flash (orchestration LLM, USD 3/M tokens, 1M-token context window), ElevenLabs Conversational AI (voice synthesis and ASR), and HiveMQ Cloud (MQTT broker, TLS-encrypted WSS) provide the AI and telemetry back-end.

5.5.2 Semantic Task Orchestrator

The orchestrator is the core engineering contribution of the software layer. Given a free-form voice command in English or Arabic — for example, “take this document to the nurses’ station and then go back to the lobby” — the orchestrator invokes Gemini 3 Flash with a structured prompt that includes the robot’s current battery level, position, known waypoints, and active task queue as live context variables. Gemini returns a mission plan as a sequence of typed steps:

$$s_i \in \{\text{navigate}(w), \text{speak}(m), \text{display}(d), \text{photo}, \text{wait}(t)\}$$

where w is a named waypoint, m is a spoken message, d is display content, and t is a wait duration in seconds. The plan is returned as a server-sent event (SSE) stream, and the mission engine begins executing the first step before the full plan has been generated. This streaming execution reduces perceived latency: the robot starts moving while the LLM is still producing step two.

The mission engine maintains a priority queue with five states: `PENDING` $\rightarrow$ `RUNNING` $\rightarrow$ `{COMPLETED, FAILED, CANCELLED}`. A 1 Hz tick evaluates the queue and dispatches navigation goals to the chassis via the `/poi` ROS service. Supported mission types are:

- **GO_TO:** Single-waypoint navigation.
- **DELIVERY:** Multi-stop sequence (up to five waypoints), with confirmation at each stop before proceeding.
- **PATROL:** Looping route through a defined set of points of interest.
- **AUTO_CHARGE:** Battery-triggered autonomous return to the docking station (configurable threshold, default 20%).

A built-in cron scheduler accepts standard five-field POSIX cron expressions, enabling recurring autonomous missions such as an hourly patrol or a nightly return-to-dock without manual intervention.

5.5.3 MQTT Bridge and Remote Dashboard

Remote operation is provided through an MQTT bridge over HiveMQ Cloud (WSS, TLS-encrypted). The bridge publishes robot state at 1 Hz: pose, battery percentage, task queue contents, navigation status, and LiDAR scan data. Large payloads (occupancy grid maps, up to several hundred kilobytes) are chunked into 64 KB segments and reassembled on the subscriber side.

The central command dashboard is a React + TypeScript single-page application. It presents a live occupancy grid map with real-time robot pose overlay, task creation interfaces (GO_TO, DELIVERY, PATROL), task queue monitoring, cron schedule management, and battery and connectivity indicators. Because it communicates exclusively over MQTT, it can be accessed from any network-connected device without VPN or special routing — a mobile phone, a laptop, or a workstation at any location with internet access.

The dashboard also provides a live telepresence view via the MJPEG stream forwarded through the MQTT bridge, allowing a remote operator to see what the robot’s camera sees while issuing navigation commands.

5.5.4 Electrical Integration

5.6 System Integration and Testing

5.6.1 Individual Unit Tests

Each subsystem was validated independently before integration:

- **Navigation:** The iMRP500 was commissioned with ROS 2 and Nav2. SLAM mapping of the test environment was completed, and waypoint navigation trials confirmed the manufacturer-specified ± 5 cm positioning accuracy. The costmap obstacle layers (LiDAR, RGBD, inflation) were verified by placing objects at known distances and confirming avoidance behaviour.
- **Sensor validation:** All sensor topics were verified over the WebSocket bridge: LiDAR scans (`/scan`), RGBD point clouds (`/upcam_data`), IMU attitude (`/imu/data`), wheel encoder odometry, battery state (`/robot_status`), and ultrasonic proximity readings. The self-diagnosis service (`/self_diagnosis`) was used to confirm all hardware subsystems reported healthy.
- **Orchestrator:** Fifteen natural-language commands of increasing complexity were submitted to the Gemini orchestrator in isolation (no robot connected), ranging from single-waypoint navigation to multi-stop conditional delivery sequences. All fifteen produced valid, executable mission plans. The streaming pipeline delivered the first step within 1.2 s of command submission on average.
- **Voice pipeline:** The ElevenLabs + Gemini 3.1 Live voice pipeline was tested for intelligibility, tool-call routing, and interruption handling (the user speaking while the robot is responding). All tool-calls (`robot_control`, `plan_and_execute`, `cancel_mission`) routed correctly to the client-side handlers.
- **MQTT telemetry:** Round-trip latency was measured by publishing 30 timestamped messages to HiveMQ Cloud and subscribing to the same topic, recording elapsed time on receipt. Results are presented in Section 5.7.
- **Body shell:** The fabricated shell was fitted to the chassis and checked against all eleven design requirements: footprint within 500 mm, total height 900 mm (within the 1,100 mm limit), mass of added structure measured and confirmed below 12 kg, display tilt verified

at 40°, storage compartment accessible and functional, no sharp edges, and removable upper tray verified.

5.6.2 System Integration Tests

With subsystems validated independently, the full system was integrated: chassis connected to tablet via Ethernet, tablet on campus WiFi, MQTT bridge active, and remote dashboard open on a separate device.

End-to-end delivery test. A delivery mission was issued by voice: the robot was instructed to navigate to a defined waypoint, announce its arrival, wait for confirmation, then return to the charging dock. The mission completed successfully across five repeated trials. The voice command was recognised, the orchestrator generated a three-step plan (navigate, speak, navigate), the mission engine executed each step, and the robot returned to dock and initiated autonomous charging.

Remote dashboard test. A navigation task was created from the remote dashboard while the robot was on the opposite side of the test space. The task appeared in the robot’s queue within one MQTT cycle (approximately 1 s), and the robot began executing within one additional tick cycle (1 Hz engine). The live map and pose overlay updated continuously, and the camera stream was viewable from the remote device.

Voice interruption test. While a patrol mission was running, a voice command was issued to cancel the current mission and go to a specific waypoint. The `cancel_mission` tool call was triggered correctly, the patrol was aborted cleanly (robot stopped, queue cleared), and the new navigation goal was dispatched within 2 s of the voice command.

5.6.3 Acceptance Test

The acceptance test scenario validates all six customer requirements (Chapter 3) in a single continuous sequence:

1. **CR1 – Autonomous navigation:** Robot navigates from charging dock to a patient room waypoint, traversing a corridor and avoiding a dynamically placed obstacle (a trolley).
2. **CR3 – Patient interaction:** On arrival, the robot greets the occupant using the voice AI, provides ward information when asked, and responds to a follow-up question in English.
3. **CR5 – User-friendly interface:** A staff member issues a delivery task via the touch display. The robot confirms the task verbally.
4. **CR1 – Multi-waypoint delivery:** Robot navigates to the pharmacy waypoint, waits for a confirmation touch, then delivers to the patient room.
5. **CR2 – Telepresence:** A remote operator connects via the MQTT-bridged dashboard and issues a navigation command while viewing the live camera stream.
6. **CR6 – Autonomous charging:** Battery level threshold triggers `AUTO_CHARGE` mission; robot navigates to dock and initiates charging.

The acceptance test was completed successfully. CR4 (infection control) is addressed at the design level through material selection (acrylic and oak-veneer surfaces compatible with hospital-grade disinfectant wipes) and is not a run-time test.

5.7 Results

5.7.1 Voice-to-Action Latency

End-to-end latency from voice command completion to first robot motion was measured across 20 trials ($N = 20$). The pipeline decomposes as:

$$\bar{t}_{v2a} = t_{\text{ASR}} + t_{\text{LLM}} + t_{\text{TTS}} + t_{\text{WS}}$$

Table 26 gives per-component statistics.

Table 26: Voice-to-action latency breakdown ($N = 20$ trials).

Component	$\bar{t}$ (ms)	σ (ms)
ElevenLabs ASR (t_{ASR})	310	45
Gemini tool-call (t_{LLM})	790	180
TTS streaming onset (t_{TTS})	400	85
WebSocket dispatch (t_{WS})	50	12
Total	1550	310

All 20 trials completed below the 3 s engineering requirement (ER3, Chapter 3), with a maximum observed latency of 2.41 s. The LLM tool-call resolution dominates at 51% of total latency. The streaming orchestrator further reduces perceived latency by pre-firing the first navigation command before the complete plan is generated, so the robot begins moving while the voice response is still being synthesised.

5.7.2 MQTT Telemetry Latency

Round-trip latency between the robot bridge and HiveMQ Cloud was measured over 30 timestamped message pairs ($N = 30$, measured 2026-03-25).

Table 27: MQTT round-trip latency statistics ($N = 30$).

Statistic	Value (ms)
Mean ($\bar{t}$)	159
Std dev (σ)	30
Median	142
95th percentile	207
Maximum	243

All messages were delivered within 243 ms. The bimodal distribution (a fast cluster at 137–157 ms and a slower cluster at 191–207 ms) reflects DPI inspection on the campus WiFi (eduroam) that introduces occasional buffering. All values are well within the limit for non-real-time telemetry; direct drive control uses the local Ethernet path and is not affected by MQTT latency.

5.7.3 Navigation Accuracy

Waypoint navigation accuracy was assessed over ten repeated trials to each of three defined waypoints. The iMRP500 positioning accuracy of ± 5 cm was confirmed in all trials. The 4-layer costmap configuration (obstacle, upward camera, downward camera, inflation) successfully avoided all statically and dynamically placed test obstacles in the acceptance test corridor without false positives causing unnecessary path deviations.

5.7.4 Orchestrator Validation

Fifteen natural-language commands of increasing complexity were submitted:

- Single-waypoint navigation (five commands): all produced correct single-step plans.
- Multi-stop delivery (five commands): all produced correctly ordered multi-step sequences with confirmation steps at intermediate waypoints.
- Conditional return (five commands, e.g., “if the battery is below 30% after delivery, go to the dock; otherwise continue patrol”): all five correctly interpreted the conditional and generated branched plans.

All fifteen commands produced valid, executable mission plans. No hallucinated waypoints or malformed step types were observed.

5.8 Standards Compliance

The following standards were applied during design and integration:

- **ISO 13482:2014** (Safety requirements for personal care robots): The 0.8 m/s maximum navigation speed and 0.8 m inflation radius in the costmap satisfy the velocity and proximity limits specified for robots operating in human-shared environments.
- **ISO 13850** (Emergency stop): The hardware emergency stop relay was tested to confirm that the robot halts within one control cycle (<100 ms) on E-stop activation. The soft-stop service (`/soft_stop`) provides a software-level equivalent.
- **ITU-T H.264** (Video compression): The MJPEG camera stream and the MQTT-forwarded video feed use standard compressed formats compatible with web browsers.
- **IEEE 802.11** (Wireless LAN): The MQTT bridge and cloud AI APIs operate over the standard campus WiFi infrastructure with no modifications to network equipment.
- **Hospital materials**: Body shell surfaces (acrylic, oak veneer) were selected to be compatible with isopropyl-alcohol-based disinfectant wipes used in clinical environments.

5.9 Budget Summary

Table 28: Hardware procurement costs (Phase 2).

Item	Cost (USD)
iMRP500 chassis (second-hand)	2,000
10-inch 1080p industrial touch display	200
Body panels, structural wood, oak veneer	800
Acrylic display frame and surround	40
Cables, fasteners, connectors, misc.	200
Total	3,240

At USD 3,240, TAMRA delivers the same functional envelope as commercial hospital AMRs priced between USD 12,000 and USD 47,500 — a 73% cost reduction compared to the nearest commercially available platform with equivalent capability. The open-source software stack (ROS 2, Nav2, standard web technologies) carries no licensing cost.

5.10 Conclusion

Chapter 5 has documented the full implementation of TAMRA, from the platform transition decision through hardware commissioning, body shell fabrication, software development, and system integration. The iMRP500 was selected after a rigorous eight-platform comparison and reverse-engineered to expose its full sensor suite over a documented WebSocket API. The body shell was designed through a three-stage workflow combining generative concept synthesis, AI-assisted 3D mesh generation, and parametric engineering refinement in Fusion 360, and fabricated locally in Oman within budget and mass constraints. The software stack — a semantic task orchestrator, voice pipeline, mission engine, MQTT bridge, and remote dashboard — was developed in parallel with the hardware procurement and completed before integration began.

Measured results confirm voice-to-action latency of 1.55 ± 0.31 s (all 20 trials below the 3 s requirement), MQTT round-trip of 159 ± 30 ms (all 30 trials below 250 ms), navigation accuracy of ± 5 cm, and 100% plan generation success across 15 orchestrator test commands. The acceptance test scenario, validating all six customer requirements in a single continuous sequence, was completed successfully. The system is realised at USD 3,240 on an open, extensible architecture.

6

Conclusions and Future Works

This chapter summarises the engineering contributions of the project, identifies areas for improvement, and charts a path for future development toward full-scale deployment in service environments.

6 Conclusions and Future Works

6.1 Project Summary

This project set out to design and build an autonomous mobile robot capable of supporting hospital operations — handling logistics, enabling remote consultation, and providing patients with a natural, conversational interface. Over two semesters, it evolved from a conceptual design exercise into a working integrated system, and in doing so required the team to make several difficult decisions that turned out to be the right ones.

The first semester produced a systematic literature survey of healthcare robotics, a formal requirements specification, a structured AHP-based system architecture selection, and a thorough grounding in ROS 2, SLAM, navigation stacks, and sensor fusion. The second semester converted that foundation into hardware and software reality.

The pivotal decision of the second semester was the platform transition. When iRobot entered bankruptcy proceedings and the iCreate 3 ecosystem became untenable, the team pivoted to a systematic evaluation of eight Chinese-manufactured mobile chassis platforms, ultimately selecting the iMRP500 — a 500 mm-diameter industrial chassis with 50 kg payload, integrated LiDAR and RGBD sensing, a 6-axis IMU, 12-hour battery life, and an onboard X86 PC running ROS 1. This decision was not a detour. The platform evaluation itself — reviewing documentation, testing ROS 2 compatibility, assessing mounting options, and benchmarking sensor suites — is one of the more thorough platform comparisons of Chinese service-robot chassis in the academic literature on affordable healthcare AMRs.

A second strategic decision widened the scope from a narrowly-framed medical device to a general-purpose autonomous service robot framework, validated in a healthcare corridor context. Medical environments impose regulatory and certification barriers that lie far outside the scope of a single academic year. Reframing TAMRA as a task-agnostic platform — capable of operating in hospitals, hospitality environments, campuses, and facilities management contexts — preserves all the technical contributions while making meaningful deployment a realistic near-term outcome.

The resulting system, TAMRA (Task-Agnostic Modular Robotic Apparatus), integrates:

- An industrial mobile chassis (iMRP500) with LiDAR SLAM and Nav2-based autonomous navigation achieving ± 5 cm positioning accuracy across mapped environments.
- A custom-designed and locally fabricated ergonomic body shell: 900 mm tall, 500 mm base footprint, oak-veneer surface finish, 40° tilted display, integrated rear storage compartment, weight under 12 kg.
- A semantic task orchestrator (Gemini 3 Flash) that converts free-form English or Arabic voice commands into structured, streaming, multi-step mission plans executed by an on-board mission engine.
- A real-time voice interface (ElevenLabs Conversational AI + Gemini 3.1 Live) with end-to-end voice-to-action latency of 1.55 ± 0.31 s ($N = 20$, all trials below 3 s).
- An MQTT-bridged remote dashboard enabling remote navigation control, live map viewing, telepresence, and fleet-ready task management from any internet-connected device.
- A dual-network tablet deployment solved through a custom ADB boot daemon, enabling simultaneous local Ethernet (robot control) and campus WiFi (cloud AI) connectivity on an Android device.

The complete system is realised at USD 3,240 — a 73% cost reduction compared to the nearest commercially available platform offering equivalent capabilities, and a fraction of the USD 12,000–USD 47,500 range occupied by commercial hospital AMRs such as the PuduBot 2 and Keenon M2.

6.2 Areas of Improvement

The implementation revealed several areas where the current system falls short of what a more mature deployment would require.

Navigation robustness in dynamic environments. The iMRP500’s navigation stack is well-calibrated for static maps. In high-traffic corridors with many simultaneous moving obstacles — people, trolleys, doors — the global planner occasionally replans inefficiently, resulting in brief stops before a new path is selected. A more adaptive dynamic obstacle layer, or an upgraded local planner, would improve fluency in busy environments.

Voice recognition accuracy in ambient noise. The ElevenLabs ASR pipeline performs well in quiet environments. In environments with background noise typical of hospital reception areas (announcements, conversations, equipment sounds), word error rates increase noticeably. Hardware-level echo cancellation (a directional microphone array) or server-side noise suppression tuned to hospital acoustic profiles would address this.

Display visibility. The 10-inch 1080p display is adequate for one-to-one interaction at close range but has limited off-axis visibility and insufficient brightness for operation near windows or in strongly lit lobbies. A higher-brightness panel, or an anti-glare coating on the acrylic frame, would improve legibility.

Autonomous docking reliability. Autonomous return to the charging dock was tested under controlled conditions. In a live deployment, small localisation drift accumulated over a long shift can cause the robot to miss the dock contact. A short-range IR or UWB beacon at the dock, used for fine-approach correction, would increase docking success rate toward the 95% engineering requirement.

Body shell durability. The wood-and-acrylic body shell meets the structural requirements for the current project timeline. For a long-term deployment, the wood subframe would need to be replaced with aluminium extrusion, and the veneer surface treatment would require periodic maintenance. The parametric Fusion 360 model makes this material substitution straightforward without dimensional changes.

Arabic language coverage. The orchestrator and voice pipeline support Arabic, but the testing conducted in this project used primarily English commands. Arabic speech recognition accuracy and orchestrator response quality in Arabic should be evaluated systematically with native speakers before Arabic is offered as a primary interaction language.

6.3 Future Works

Multi-robot fleet coordination. The MQTT bridge architecture supports multiple robots publishing to distinct topic namespaces from a single dashboard. The next step is to implement a fleet manager that can distribute tasks across a pool of TAMRA units, prevent waypoint conflicts, and rebalance loads when one robot goes to charge. ROS 2’s DDS-based distributed computing layer provides the underlying communication primitives for this.

Scheduled and recurring missions. The cron scheduler in the current mission engine supports POSIX five-field expressions and has been tested for simple recurring patterns (hourly patrol, nightly dock return). Extending this to more complex temporal constraints — “deliver medications at 08:00, 14:00, and 20:00 on weekdays, skipping public holidays” — requires calendar integration and exception handling logic that is a natural near-term extension.

Hospital Information System integration. Connecting TAMRA to a hospital’s HIS or electronic health record (EHR) system would allow the orchestrator to read patient room assignments, scheduled deliveries, and ward status dynamically, rather than relying on manually configured waypoints. This integration requires compliance with HL7/FHIR data standards and appropriate access controls, and represents the primary path toward clinical deployment.

Emotional intelligence and affective response. The current voice interface is transactional: the robot completes tasks and responds to queries. Adding affective computing — detecting stress or distress in the patient’s voice through prosody analysis, or recognising facial expressions via the RGBD camera — would allow the robot to adapt its tone and responses to the emotional state of the person it is interacting with. This is particularly valuable for paediatric and elderly patient populations.

UV-C disinfection module. The storage compartment and upper tray mounting points provide a structural basis for adding a UV-C emitter module for autonomous disinfection runs during off-peak hours. UV-C integration would require safety interlocks (the emitter must be disabled when people are detected nearby) and scheduling logic that coordinates with the patrol mission type.

UWB docking assistance. Ultra-wideband ranging at the charging dock would provide centimetre-level approach guidance independent of LiDAR localisation drift. A single UWB anchor at the dock and a corresponding tag on the chassis base would correct the final approach and improve docking success rate to the 95% engineering target under all operating conditions.

Longitudinal field study. The most important future work is deployment in a real operating environment. A structured field study in a hospital, hotel, or campus building — measuring staff workload reduction, patient interaction quality, navigation reliability over multi-week operation, and maintenance burden — would provide the empirical grounding that laboratory bench tests cannot. The open architecture and low hardware cost of TAMRA make it viable to deploy multiple units for such a study without the procurement barriers associated with commercial platforms.

Generalisation to other service contexts. TAMRA was validated in a healthcare corridor. The same hardware and software stack applies directly to hotel concierge services (delivering items to rooms, responding to guest questions), airport wayfinding, library book retrieval, and facilities management. Each context requires a different waypoint map and orchestrator prompt configuration but no architectural changes. Demonstrating the platform across multiple service contexts would substantiate the task-agnostic design claim.

6.4 Closing Remarks

Service robotics is at a turning point. The hardware required to build a capable autonomous mobile robot has become affordable. The AI required to give that robot a natural voice and a reasoning capability has become accessible. What remains difficult is integration — combining navigation, voice, task management, network communication, and physical design into a system that is reliable enough to be genuinely useful.

TAMRA demonstrates that this integration is achievable by a two-person undergraduate team, at a cost that makes it realistic for hospitals, businesses, and institutions that cannot afford commercial alternatives. The platform is open, documented, and extensible. Every engineering decision made during this project — the platform selection, the material choices, the software

architecture, the network routing solution — is recorded in this report so that the next team, the next institution, or the next generation of this system can build on it rather than repeating it.

The first semester built the knowledge. The second semester built the machine.

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